澳門理工大學
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AIHA8122 · FRONTIERS IN SMART HEALTHY AGEING

Class 6: Machine learning and Deep learning

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Small Sample Size

What Does 5-Fold Cross-Validation Mean?

Train on part of the dataset; evaluate on observations excluded from that model fit.

1. **Divide the dataset into 5 folds**
   Five approximately equal, non-overlapping groups.
2. **Train on 4; evaluate on 1**
   About 80% for training and 20% for evaluation.
3. **Repeat until every fold is held out**
   Five model fits; each observation receives one held-out prediction per CV run.
4. **Summarize predictive performance**
   For example, summarize classification accuracy across the held-out predictions.

Example: 50 observations → 5 folds of 10

Fit	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
1	Evaluate	Train	Train	Train	Train
2	Train	Evaluate	Train	Train	Train
3	Train	Train	Evaluate	Train	Train
4	Train	Train	Train	Evaluate	Train
5	Train	Train	Train	Train	Evaluate

Training: 40 observations Evaluation: 10 observations

For classification
Use stratified folds to approximately preserve class proportions.

For repeated measurements
Keep all observations from the same patient in the same fold.

Avoid leakage: Fit scaling and feature selection using training observations only—not the full dataset.

Key idea: Cross-validation reuses limited data efficiently; it does not create new independent observations.

Small Sample Size

5-Fold, 10-Fold, or Leave-One-Out?

More folds mean more training data per fit— but not automatically a better performance estimate.

Example: a dataset with 50 observations

Method	Training per fit	Evaluation per fit	Number of fits*	Main trade-off
5-fold CV	40 observations	10 observations	5	Larger evaluation folds; fewer model fits.
10-fold CV	45 observations	5 observations	10	More training data; smaller evaluation folds.
LOOCV Leave-one-out CV	49 observations	1 observation	50	Nearly all data used for training; many highly overlapping training sets.

*For one fixed model configuration; tuning and repeated CV require additional fits. In LOOCV, hold out each observation in turn and combine the 50 held-out predictions.

When to consider 5- or 10-fold CV

Common choices for predictive evaluation. For classification, ensure evaluation folds contain enough observations from each class. Repeated splits can reveal sensitivity to the partition.

When to consider LOOCV

When retaining almost all observations for training is important and fitting is affordable. It is not automatically best for small samples: prediction-error estimates can have high variance.

LOOCV metric caution: A single held-out observation cannot support a per-fold AUC. Combine held-out predictions for an overall assessment; do not average single-observation AUCs.

Practical starting point: Consider stratified 5-fold CV for classification, but reduce the fold count when a class is too small. No fold count fixes an inadequate sample.

Small Sample Size

Cross-Validation or Bootstrap?

Choose the resampling method according to the question.

Cross-validation

1. **Partition into held-out folds**
   Training and evaluation observations are separate within each model fit.
2. **Common use: prediction and tuning**
   "How well will this model predict observations it has not seen?"
3. **PLS-DA example**
   Select component count in inner CV; estimate predictive performance in outer CV.

Separate tuning from evaluation

Use nested CV—or an untouched test set— when model settings are selected from the data.

Bootstrap

1. **Sample with replacement**
   Usually draw n observations from the original n; some are duplicated and others omitted.
2. **Common use: uncertainty and stability**
   Estimate sampling variability, standard errors, or confidence intervals when appropriate.

Illustration: 6 original observations

Original IDs: 1, 2, 3, 4, 5, 6

2	2	4	5	5	6

One bootstrap sample: IDs 2 and 5 repeat; IDs 1 and 3 are omitted.

Recalculate the statistic in many resamples. Bootstrap confidence intervals can be unreliable for very small or degenerate samples.

Not interchangeable: Bootstrap can also assess prediction using dedicated validation procedures. Scoring a model on its bootstrap training sample is not a held-out performance estimate.

CV: "Will it predict?" **Bootstrap:** "How uncertain is the estimate?"
Neither method replaces additional independent data.

Class Imbalance

When the event of interest is much less common than its absence.

01 · CLASS DISTRIBUTION

Majority and minority classes

Some outcome classes have **many more examples** than others.



Each dot represents one patient

95
Majority
No event · y = 0

5
Minority
Event · y = 1

Illustrative example: a cohort of 100 patients with 5 events.

02 · EVALUATION PITFALL

Accuracy can be misleading

Predict **"no event"** for every patient.

95% **accuracy**

Accuracy = correct / total
= 95 / 100

All 5 events are missed.
Sensitivity = 0 / 5 = 0%

High overall accuracy does not guarantee that the model detects events.

03 · CLINICAL IMPACT

Clinical costs may be asymmetric

False negative: missed event
Predict "no event" when the event occurs.
Possible harm: delayed or missed care.

False positive: false alarm
Predict "event" when no event occurs.
Possible harm: unnecessary testing or treatment.

Cost(FN) may differ from Cost(FP)

Which error matters more depends on the clinical task and resulting action.

Class Imbalance

Three different levers: the training loss, the decision threshold, and the ensemble design.

01 · CHANGE THE TRAINING LOSS

Class weights

Give minority-class examples **more influence during training**, without adding or deleting patients.

$$L = (1/N) \Sigma w_{y_i} \ell(y_i, p_i)$$

ℓ = prediction loss; w = class-specific weight.

Example: 95 non-events, 5 events

Inverse-frequency weighting gives $w_1 / w_0 = 19$.

Each event contributes 19 times the weight of each non-event.

Frequency-based weights are not automatically clinical harm costs. Tune weights on

02 · CHANGE THE DECISION RULE

Threshold adjustment

Convert a predicted event probability into an alert using a threshold **t**.

Alert if p(event | x) ≥ t

t = 0.50
For p = 0.20:
No alert

t = 0.10
For p = 0.20:
Alert

Lowering t generally detects more events, but also produces more false alarms.

Clinical example: choose a threshold to meet a prespecified sensitivity target, then examine false alarms and workload.

03 · CHANGE THE ENSEMBLE DESIGN

Imbalance-aware ensembles

Combine models with **explicit weighting or resampling**.

Balanced Random Forest

Train trees using balanced resampled subsets, then combine their predictions.

EasyEnsemble

Combine boosted models trained on different balanced subsets.

Weighted gradient boosting

Use class/sample weights; for example, XGBoost offers `scale_pos_weight`.

Standard Random Forest, AdaBoost, and XGBoost are not inherently imbalance-proof.

XGBoost: scale_pos_weight

Directly adjust the loss function

$$\text{scale_pos_weight} = \frac{\text{Negative Count}}{\text{Positive Count}}$$

Class Imbalance

Under-sampling removes examples; over-sampling repeats or creates examples.

01 · REMOVE MAJORITY EXAMPLES

Under-sampling

Random: randomly retain a subset of majority-class patients.

NearMiss-2

For each majority example, calculate its average distance to the **k farthest minority examples**.

Keep majority examples with the smallest averages. “Farthest” distinguishes version 2.

02 · REPEAT MINORITY EXAMPLES

Random over-sampling

Sample minority-class patients **with replacement**.

A, B, C → A, B, C, A, C, B …

More training rows,
but no new independent patients.

A simple comparator that does not require feature-space interpolation.

03 · CREATE MINORITY EXAMPLES

SMOTE

Interpolate between a minority example and a nearby minority neighbor.

$$x_{new} = x_i + \lambda(x_j - x_i)$$

$0 \le \lambda \le 1$



Can propagate noise or create implausible clinical

Beyond standard SMOTE: match the method to the data

SMOTENC

For mixed numerical and categorical features. Avoid ordinary interpolation of category codes.

Borderline-SMOTE

Focuses synthesis on minority examples near class boundaries. It is not a guaranteed noise-removal method.

ADASYN

Generates more examples in difficult local regions. Inspect whether those regions reflect noise.

SMOTE + cleaning

SMOTE-Tomek or SMOTE-ENN combines synthesis with removal of selected overlapping or inconsistent examples.

Class Imbalance

Rule 1

Mild or moderate data imbalance cases (e.g., minority-to-majority ratio is equal to or less than 1 : 10)

- **Simpler techniques such as oversampling or undersampling can be considered**.
- For datasets with relatively small sizes and low dimensinality, **oversampling like SMOTE can effectively enhance minority class representation without significantly increasing complexity**.

Rule 2

If the dataset is large

- **undersampling techniques** often become the method of choice due to their computational efficiency.
- However, undersampling should be performed carefully to avoid discarding valuable information.
- **Techniques such as Tomek Links** are recommended to selectively remove redundant majority class samples while preserving important data points.

Rule 3

In scenarios with severe imbalance (e.g., minority-to-majority ratio of 1 : 50 or higher

- Traditional sampling methods alone may be insufficient.
- Leveraging more sophisticated approaches such **as deep generative models like GANs** becomes advantageous.

Class Imbalance

Change the training data—not the population used to evaluate the model.

Split first → resample training only

Original patient data

Split by patient, time, or site as required by the study design.

→

Training fold

Fit preprocessing → apply SMOTE or other resampling → fit the model.

Validation fold

Do not rebalance.
Use to compare methods and tune settings.

Apply preprocessing learned from the training fold.

Final test set

Do not rebalance.
Keep separate from tuning.
Evaluate the final model.

Preserve the intended clinical population.

Why "SMOTE first" is wrong

Entire dataset → SMOTE → split into folds ×

Synthetic training examples may be constructed using information from patients later assigned to validation or test data.

Result: information leakage and potentially overly optimistic performance.

Clinical example: preserve event prevalence

5%

Intended population:
50 events per 1,000 patients

≠

50%

Artificially balanced test:
not the intended population

Rebalancing evaluation data changes the question being answered. Precision and clinical alert burden must be assessed under realistic prevalence.

Sources: imbalanced-learn, Common pitfalls; scikit-learn, Cross-validation · Example is illustrative.

The Confusion Matrix

Compare predicted labels with actual outcomes: four possible results for every patient.

Predicted label → / Actual outcome ↓	Predicted positive	Predicted negative
Actual positive — Condition present	**TP True positive** — The model predicts the condition, and the condition is present. **Correctly detected case**	**FN False negative** — The model predicts no condition, but the condition is present. **Missed case**
Actual negative — Condition absent	**FP False positive** — The model predicts the condition, but the condition is absent. **False alarm**	**TN True negative** — The model predicts no condition, and the condition is absent. **Correctly identified non-case**

True / false = was the prediction correct?
True means correct; false means incorrect.

Positive / negative = what did the model predict?
Positive means the defined outcome of interest is present.

Clinical Meaning of Classification Errors

A missed case and a false alarm can have very different consequences.

FN False negative

Missed disease or deterioration

Actual outcome: positive
Model prediction: negative

Possible consequences: false reassurance, delayed assessment, or a missed opportunity to intervene.

FP False positive

False alarm

Actual outcome: negative
Model prediction: positive

Possible consequences: unnecessary testing or treatment, anxiety, and additional resource use.

Error costs depend on context

Consider disease severity, urgency, and what happens after a positive prediction.

- How harmful is a missed case?
- How invasive or risky is the follow-up action?
- Can the service manage additional false alarms?

No metric is universally best

Different metrics answer different questions. Choose measures that match the intended clinical use, and assess benefits alongside harms.

Start with the clinical decision—not with a favorite metric.

Evaluate the consequences of errors, not just how often they occur.

Core Classification Metrics

Same confusion matrix—different questions about performance.

1 · Identify actual cases and non-cases

Sensitivity = TP / (TP + FN)

Among actual cases, how many are detected?

Specificity = TN / (TN + FP)

Among actual non-cases, how many are ruled negative?

Sensitivity is also called recall or the true-positive rate.

2 · Interpret the model's predictions

PPV = TP / (TP + FP)

Among positive predictions, how many are actual cases?

NPV = TN / (TN + FN)

Among negative predictions, how many are actual non-cases?

PPV = precision. Both PPV and NPV depend on prevalence.

3 · Summarize correctness across classes

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Proportion of all predictions that are correct.

Balanced accuracy = (Sensitivity + Specificity) / 2

Gives equal weight to performance in each actual class.

Accuracy can look high when the majority class dominates.

4 · Combine performance into one score

F1 = 2TP / (2TP + FP + FN)

Harmonic mean of precision and recall; excludes TN.

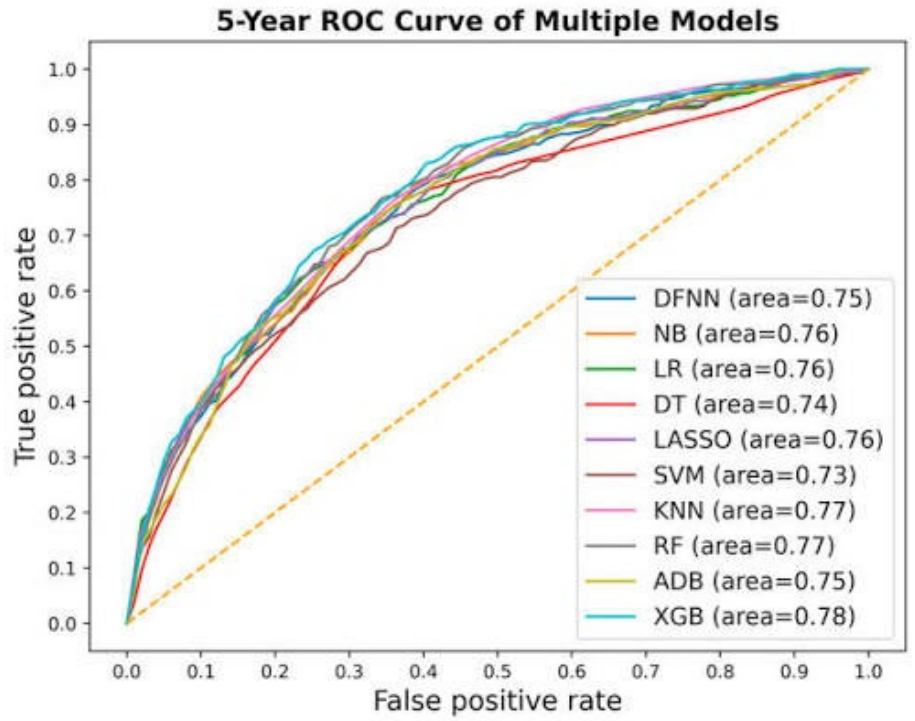
$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{[(TP + FP)(TP + FN)(TN + FP)(TN + FN)]}}$$

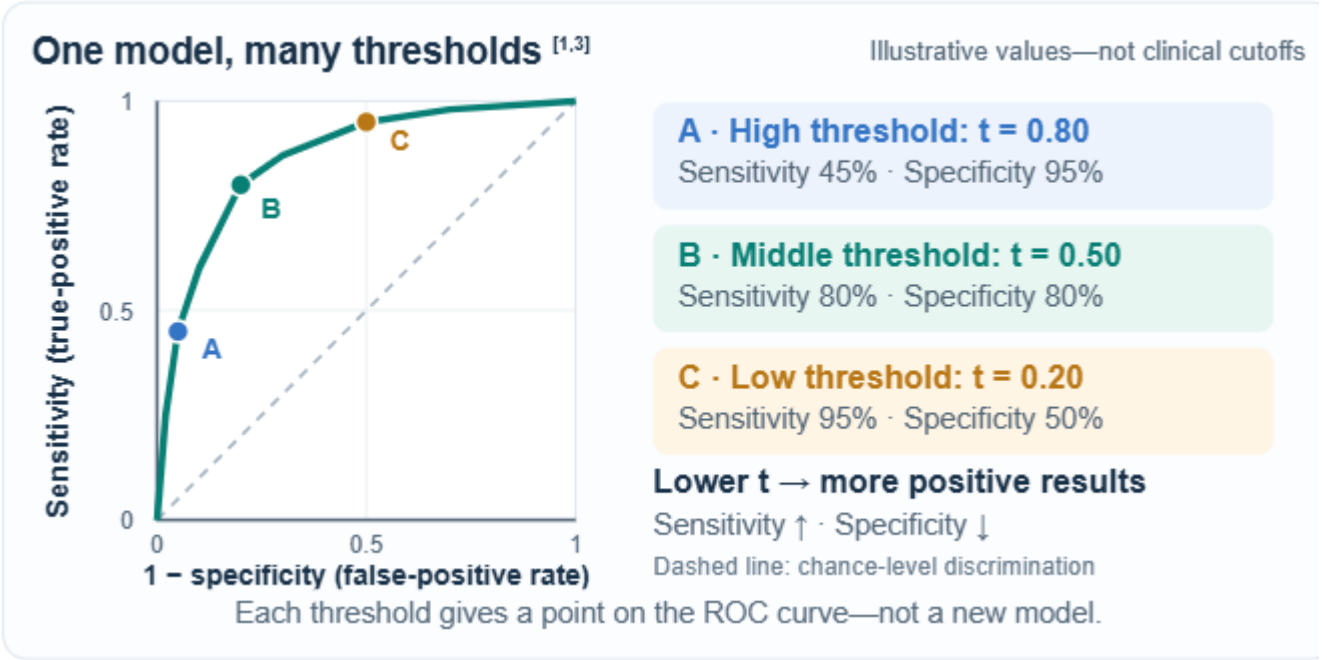
Uses all four counts: +1 perfect agreement; 0 no correlation; −1 complete disagreement.

Select metrics that answer the clinical question—not just the highest score.

Receiver Operating Characteristic (ROC) curve

A Receiver Operating Characteristic (ROC) curve is a graphical plot that evaluates the performance of a binary classification model or diagnostic test across **all possible classification thresholds**





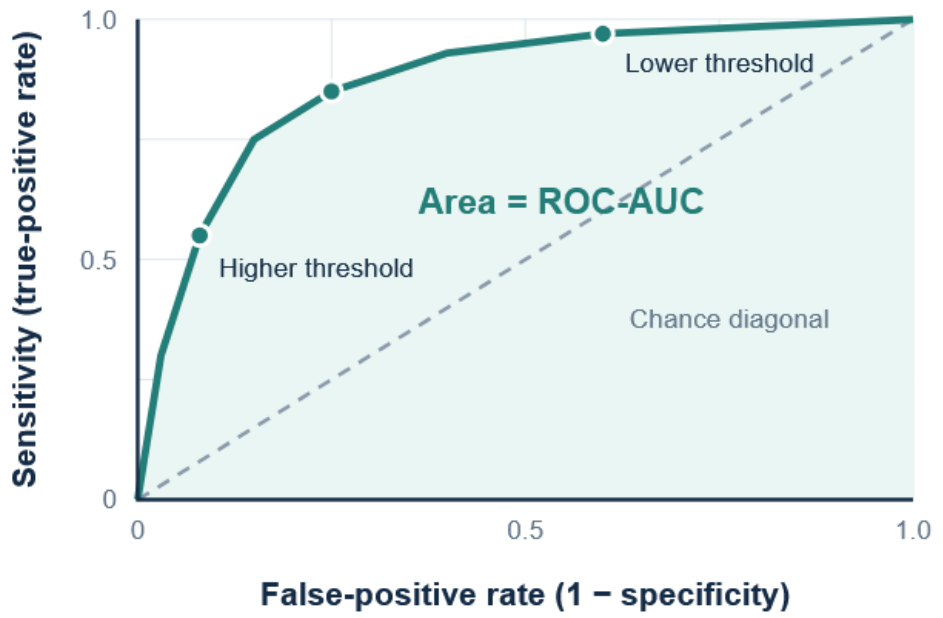
Do we usually use 50%?
0.50 is a common software default for probability outputs—not a universal clinical cutoff. ROC curves can also use scores that are not probabilities.

Threshold: The cutoff that converts a score into a positive or negative result. Predict positive if score ≥ t

ROC Curve and ROC-AUC

How well does the model separate cases from non-cases?

The ROC curve



Illustrative curve—not study results. Each point corresponds to a classification threshold.

Performance across thresholds

Lowering the threshold generally detects more cases, but also produces more false positives.

Sensitivity = TP / (TP + FN) · FPR = FP / (FP + TN)

ROC-AUC measures discrimination

Summarizes how well model scores rank cases above non-cases across thresholds.

0.5 · Chance-level discrimination

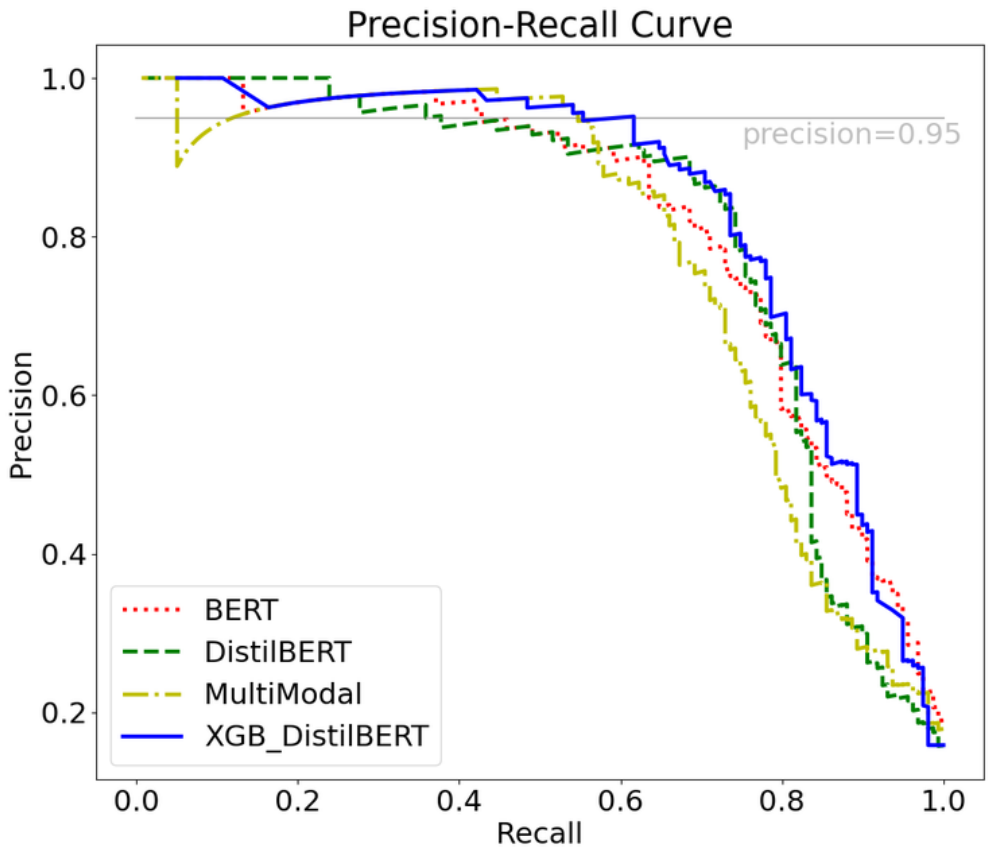
1.0 · Perfect discrimination

Not a measure of calibration

A high ROC-AUC does not show that predicted risks are accurate. Calibration asks whether predicted probabilities match observed outcome frequencies.

Discrimination: who is at higher risk? Calibration: how accurate is the estimated risk?

Precision–Recall (PR) curve



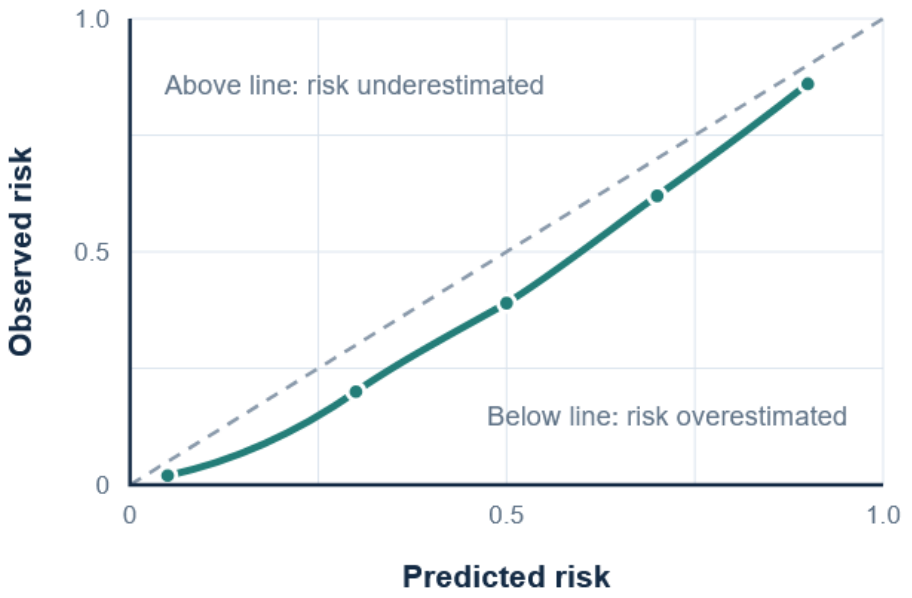
When should we provide it?
Add a PR curve alongside ROC when disease is rare or positive alerts trigger costly follow-up. It makes the false-positive burden easier to see.

How do we read it?
Top-right is ideal. At the recall required clinically, read the precision: how trustworthy are the positive results at that operating point?

Calibration

Do predicted risks agree with observed outcome frequencies?

The calibration plot



Dashed line = ideal agreement · Teal curve = illustrative model. Not study results.

Calibration intercept **Ideal: 0**

Assesses systematic over- or underprediction: positive = risks too low; negative = risks too high.

Here, intercept means calibration-in-the-large: estimated with the calibration slope fixed at 1.

Calibration slope **Ideal: 1**

<1: predictions are too extreme.
>1: predictions are not extreme enough.

Estimated by regressing the binary outcome on the log-odds of predicted risk, with a free intercept.

Brier score

Brier = (1/n) Σ $(p_i - y_i)^2$

Mean squared probability error; lower is better.
p_i = predicted risk; y_i = outcome (0 or 1).

An overall performance measure—not calibration alone.

The lower is better [0 – 1]
Usually should be below 0.20, close to 0

A predicted risk of 20% should correspond to about 20% observed events among similar-risk patients.

Four ROC Pitfalls—and How to Avoid Them

A practical checklist before presenting your results

1. ~~**Treat training-set AUC as the final result**~~
   Validate on unseen data: use a held-out test set; add external validation when possible.

2. ~~**Report only ROC AUC when the outcome is rare**~~
   Add a PR curve and report prevalence; assess calibration if predicted risks will guide decisions.

3. ~~**Assume the higher AUC means a better model**~~
   Compare on the same test population: report uncertainty in the difference and consider clinical value.

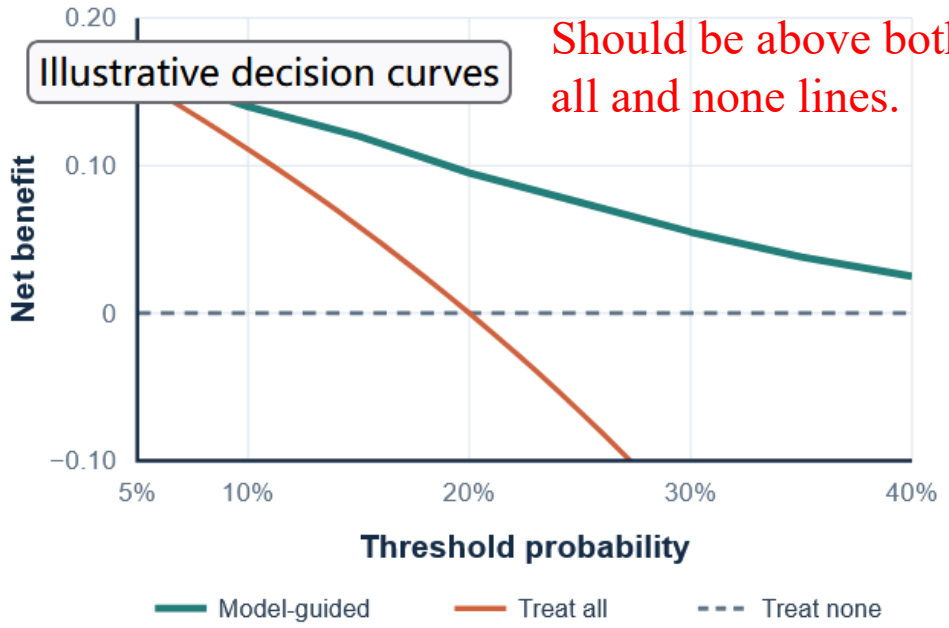
4. ~~**Leave the population and outcome unclear**~~
   Define who, what, and when: state the target population, positive outcome, and prediction horizon.

> **Before you present:** Was it validated? Is AUC enough? Is the difference meaningful? Who does the result apply to?

Decision Curve Analysis

Does using the model improve the balance of benefits and harms?

Compare decision strategies



Illustrative curves—not study results. Interpret only over clinically plausible thresholds.

Threshold probability (p_t)

The risk at which action becomes worthwhile. Act when predicted risk is at or above p_t.

Reflects the trade-off between appropriate action and unnecessary intervention.

Net benefit

$$NB = TP/n - (FP/n) \times p_t/(1 - p_t)$$

Credits true-positive decisions and penalizes false-positive decisions according to the threshold.

TP = true positives; FP = false positives; n = sample size. At p_t = 20%, each false

Connect prediction to action

Higher net benefit is better at a given threshold. Model use adds value when it exceeds both treat-all and treat-none.

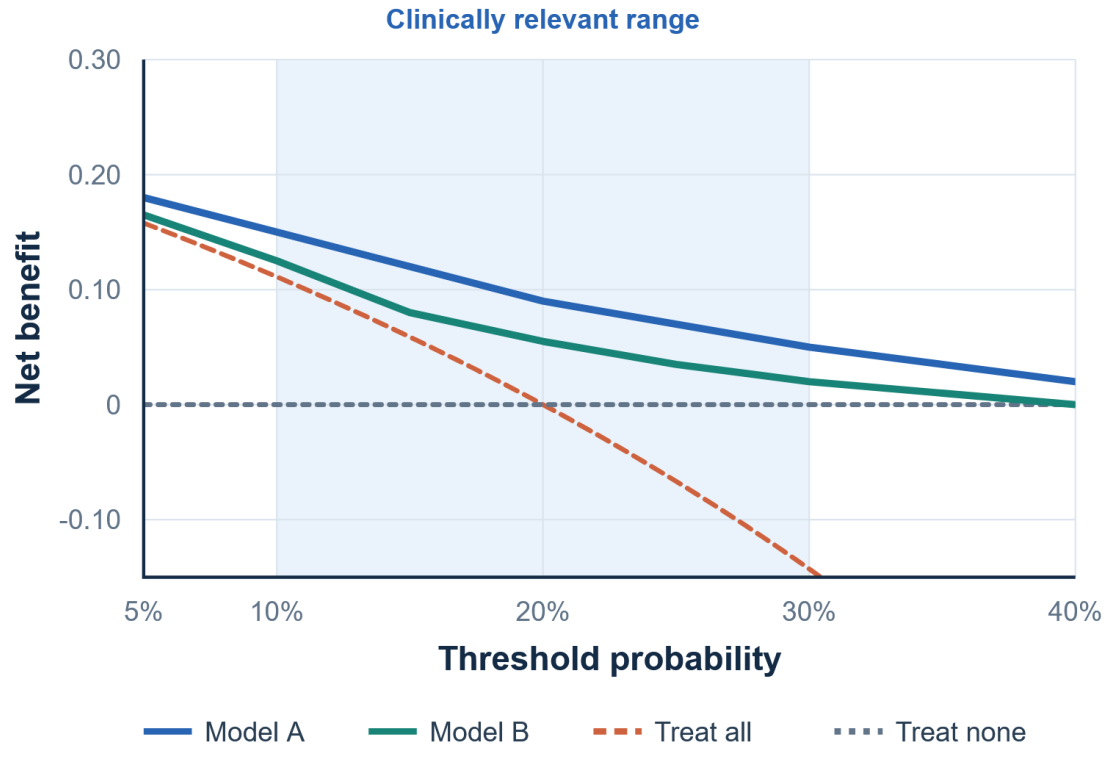
"Treat" can mean treatment, testing, or referral. DCA estimates utility—not proof of improved outcomes.

A useful model must improve decisions—not just discrimination.

Decision Curve Analysis

Which strategy is preferred based on these curves?

A referral pathway uses threshold probabilities of **10%–30%**. Compare the strategies within the shaded range.



Shading = clinically relevant thresholds. Synthetic teaching curves; assumed event prevalence = 20%.

A. Model A: it has the highest net benefit throughout 10%–30%.

B. Model B: a lower curve means fewer unnecessary referrals and therefore better utility.

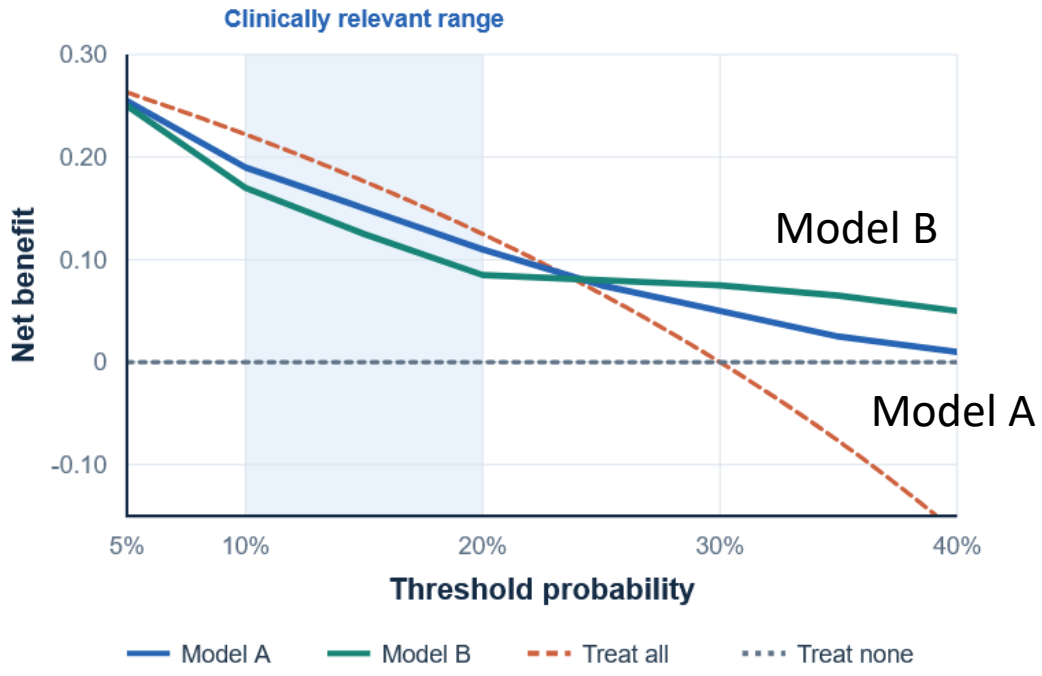
C. Treat all: this guarantees that every patient with the outcome is referred.

D. Treat none: zero net benefit is always the safest choice.

Decision Curve Analysis

Does the better model add clinical value?

For this intervention, only thresholds of **10%–20%** are clinically acceptable. Both models perform better at some higher thresholds.



Shading = clinically relevant thresholds. Synthetic teaching curves; assumed event prevalence = 30%.

Which interpretation is most appropriate?

A. Use Model A because it is better than Model B within 10%–20%.

B. Use Model B because it performs best at thresholds above 25%.

C. Neither model adds net benefit over treat-all within 10%–20%.

D. Use either model because both have positive net benefit within 10%–20%.

Regression Metrics

For continuous outcomes: how wrong are predictions, and how uncertain are they?

Mean absolute error — MAE

↓ Lower is better

$$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|$$

Average absolute prediction error, expressed in the outcome's units.

Example: MAE = 2 days means an average absolute error of 2 days—not that every prediction is within 2 days.

MSE and RMSE

↓ Lower is better

$$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2 \quad \cdot \quad RMSE = \sqrt{MSE}$$

Squared errors give large misses more influence.

MSE uses squared units; RMSE uses the original units. Useful when large prediction errors are especially costly.

R-squared — R^2

↑ Higher is better

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$$

Compares squared error with predicting the evaluation-set mean.

1 = perfect; 0 = mean-baseline performance; < 0 = worse than baseline. Not "percent accuracy." Assumes nonconstant outcomes.

Prediction intervals

Individual uncertainty

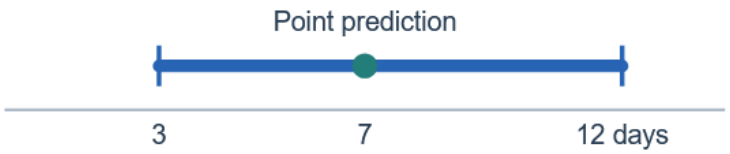
Point prediction + lower and upper bounds

A range for a new individual outcome, rather than just an estimated value.

A valid 95% interval procedure targets 95% coverage over repeated predictions under its assumptions. Inspect coverage and width—not width alone.

Illustration: hospital stay

Predicted stay: **7 days**
95% prediction interval: **3–12 days**



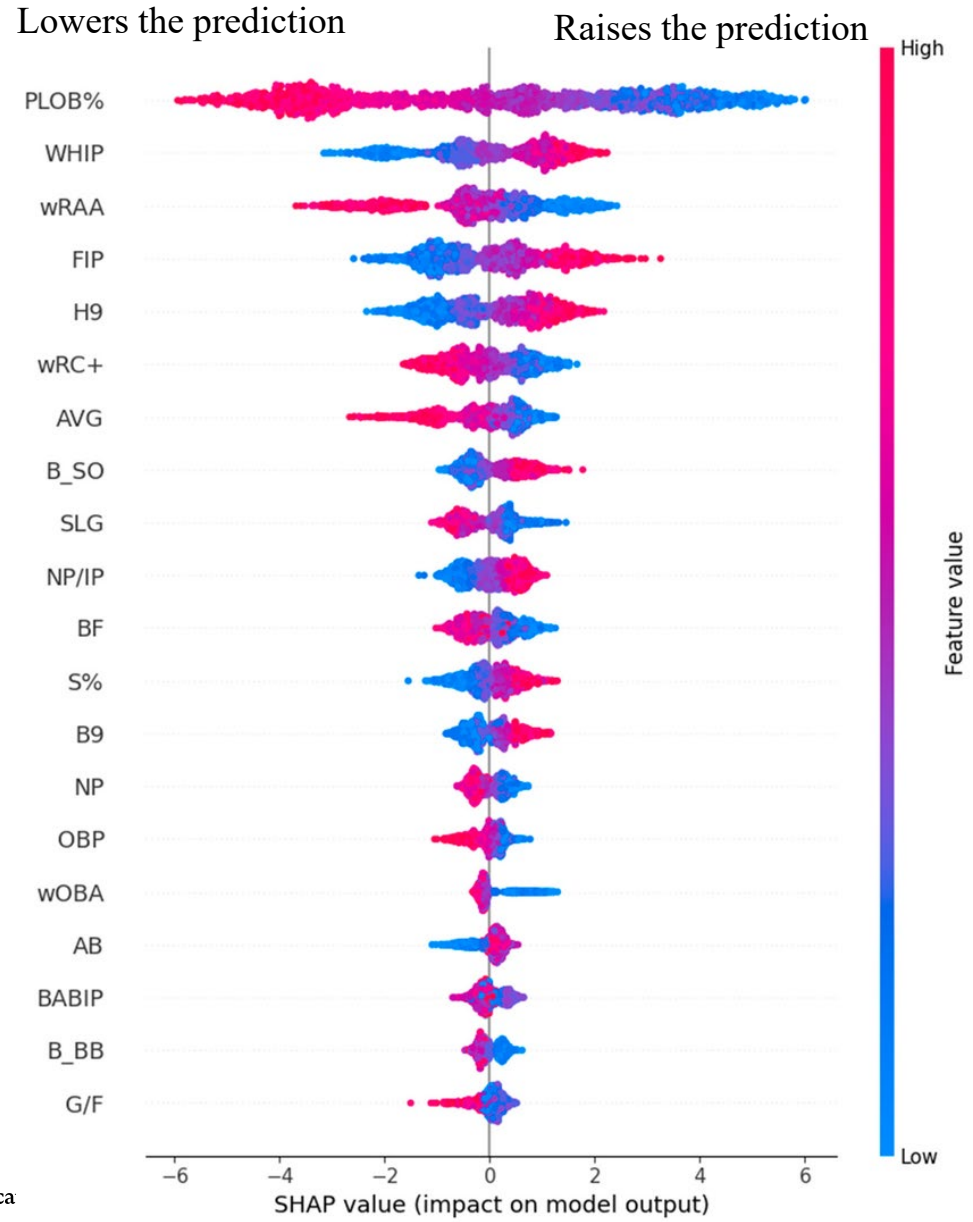
Same point estimate—important uncertainty for planning.

Synthetic example, not study results. A prediction interval is not a confidence interval for the mean.

Report error in meaningful units—and show uncertainty around individual predictions.

SHAP (SHapley Additive exPlanations)

How to Read a SHAP Summary Plot



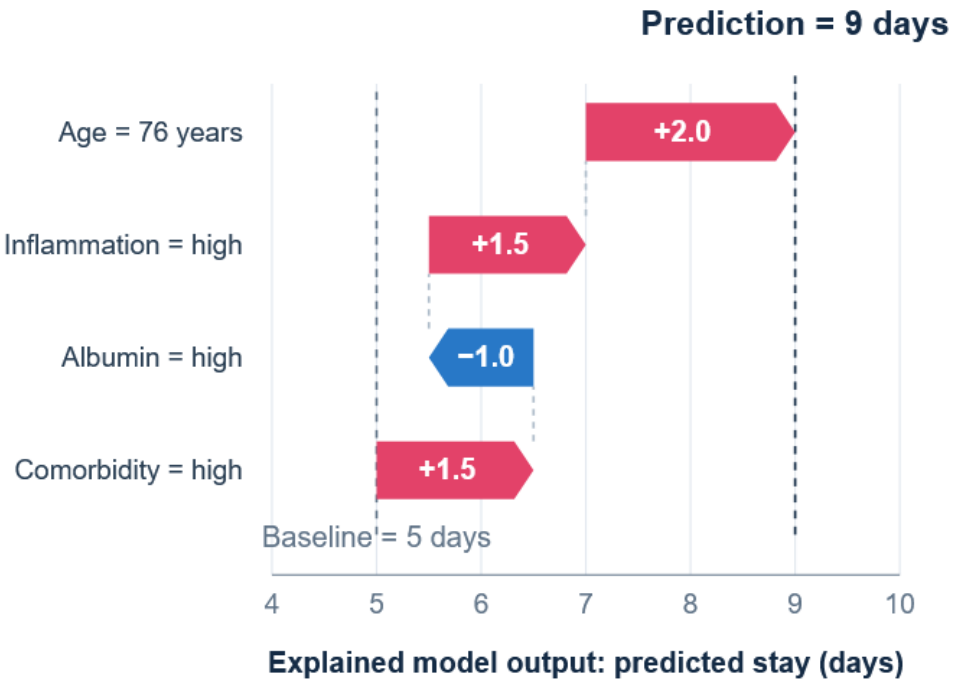
Four things to read

- **Rows: features, typically ranked by average absolute SHAP contribution. (Important ones are on the top)**
- Each dot: one patient's contribution for that feature.
- **Horizontal position**: left lowers the prediction; right raises it. Farther from zero means a larger contribution.
- Color: red **= high feature value**; blue = low feature value**. Color alone does not tell you the direction.**

Across patients: Which features influence predicted hospital stay? (Feature importance ranking

SHAP (SHapley Additive exPlanations)

How to Read a SHAP Waterfall Plot



Synthetic teaching example. Read from the bottom upward. All four feature contributions are shown.

One patient: How does the model move from a baseline of 5 days to a prediction of 9 days?

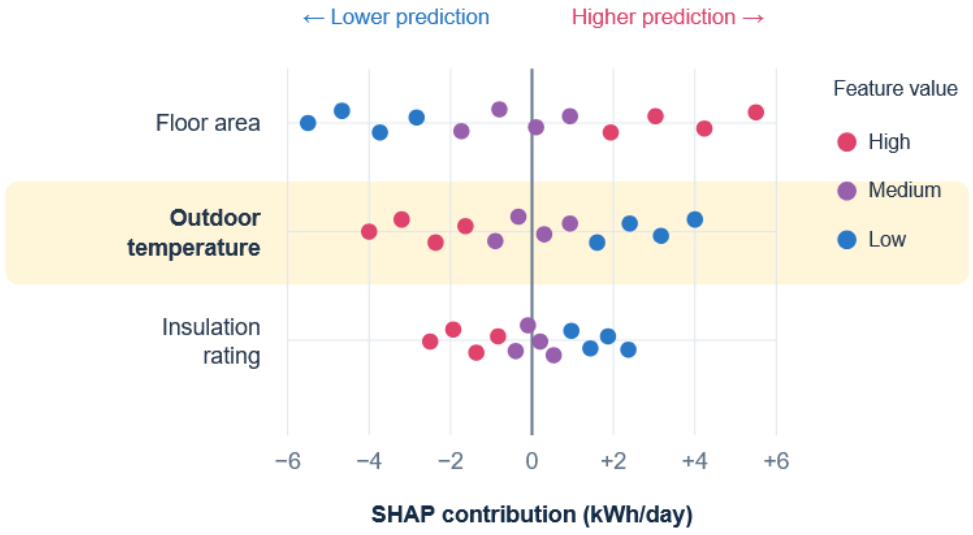
Follow the contributions

1. **Start at the baseline:** the expected model output for the selected background/reference data.
2. **Add each contribution:** red raises the prediction; blue lowers the prediction.
3. **Reach the final prediction:** 5 + 1.5 − 1.0 + 1.5 + 2.0 = **9 days**.
4. **Check the units:** these contributions are in days. Other SHAP figures may explain probability or log-odds instead.

SHAP (SHapley Additive exPlanations)

A model predicts daily household electricity use. The plot shows feature contributions across households.

Contributions to predicted electricity use



Synthetic teaching example. Each dot represents one household's contribution for a feature. Rows are ordered by mean absolute SHAP contribution.

In the highlighted temperature row, most red dots are left of zero. Which interpretation is correct?

A Higher outdoor temperatures generally contribute to higher predicted electricity use.

B Higher outdoor temperatures generally contribute to lower predicted electricity use.

C Red dots represent households with low outdoor temperatures.

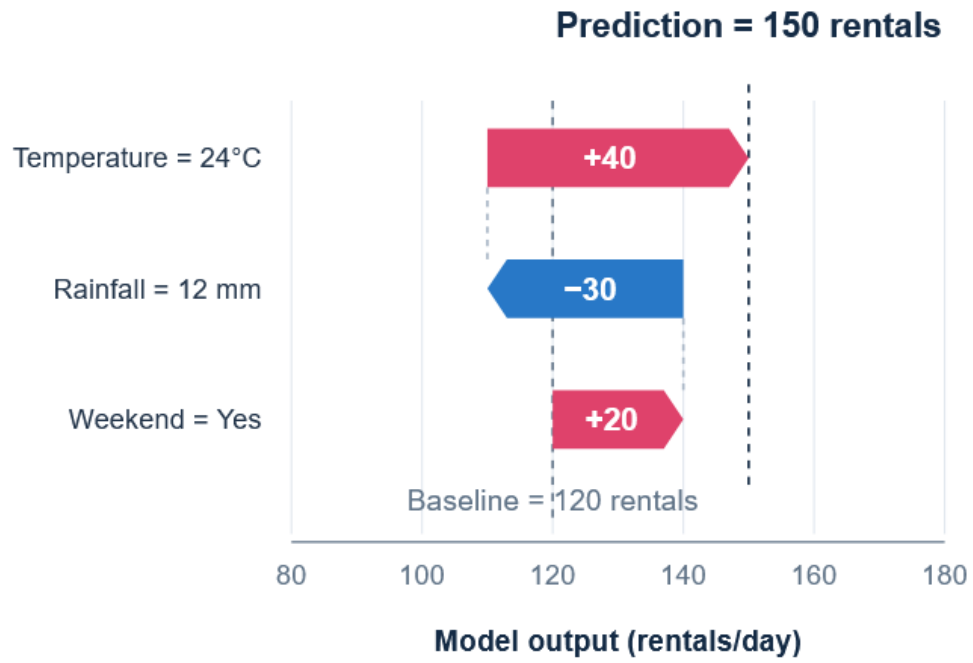
D The plot proves that increasing outdoor temperature causes electricity use to decrease.

SHAP (SHapley Additive exPlanations)

From Baseline to Prediction

A model predicts daily bicycle rentals. This waterfall explains the prediction for one day.

Contributions for one day



Synthetic teaching example. Read from the bottom upward. All feature contributions are shown. Red raises the prediction; blue lowers it.

Which statement correctly explains the rainfall contribution and the final prediction?

A Rainfall lowers the prediction by 30 rentals. The net contribution is +30, giving a prediction of 150 rentals.

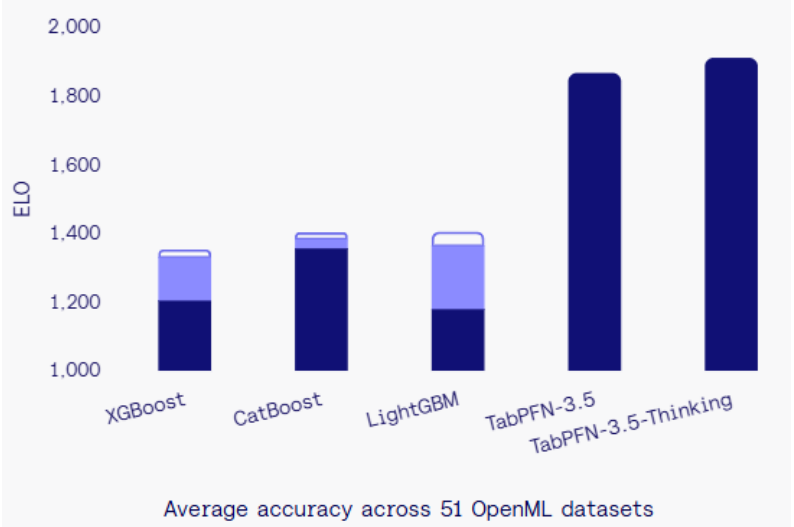
B Rainfall raises the prediction by 30 rentals. The final prediction is 210 rentals.

C Rainfall lowers the prediction by 30 rentals. The final prediction is 30 rentals because that is the sum of the contributions.

D Rainfall has no effect on the prediction because its contribution is negative.

Tabular Foundation Model

- The tabular foundation model for state-of-the-art predictions on structured data.



nature › articles › article

Article | Open access | Published: 08 January 2025

Accurate predictions on small data with a tabular foundation model

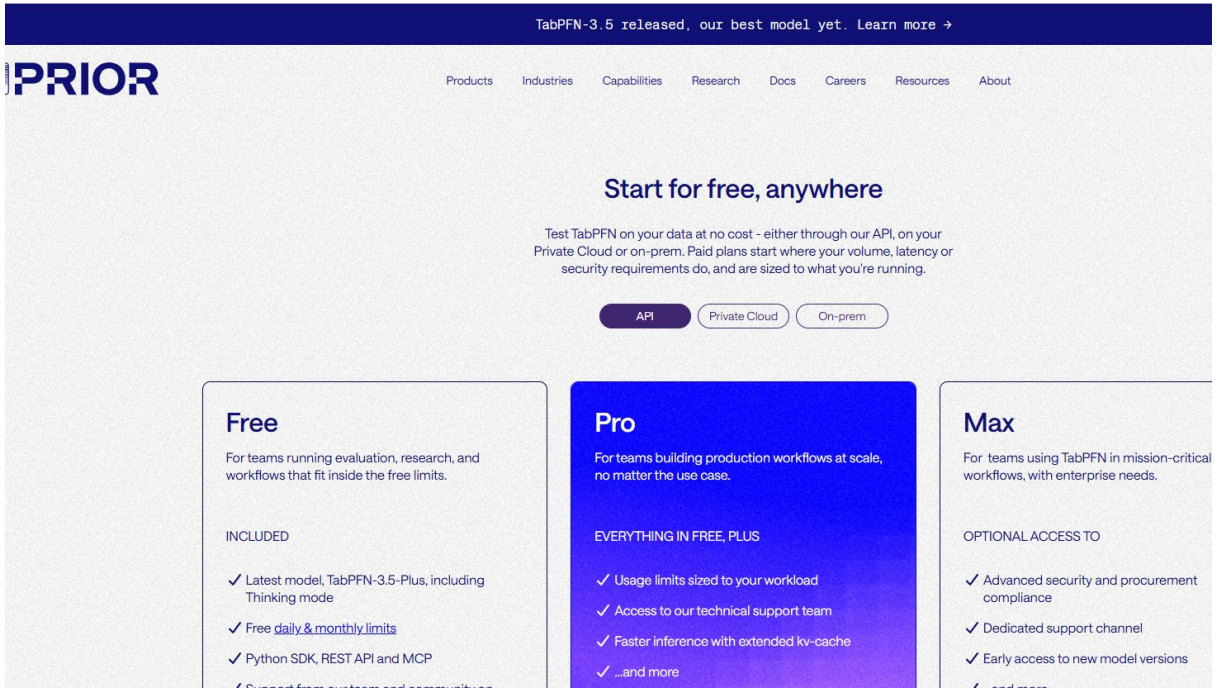
Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister & Frank Hutter

Nature **637**, 319–326 (2025) | Cite this article

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https://priorlabs.ai/pricing



Principal Component Analysis (PCA)-Unsupervised

1. **Unsupervised linear transformation**

   Components are linear combinations of the original features.

2. **New orthogonal components**

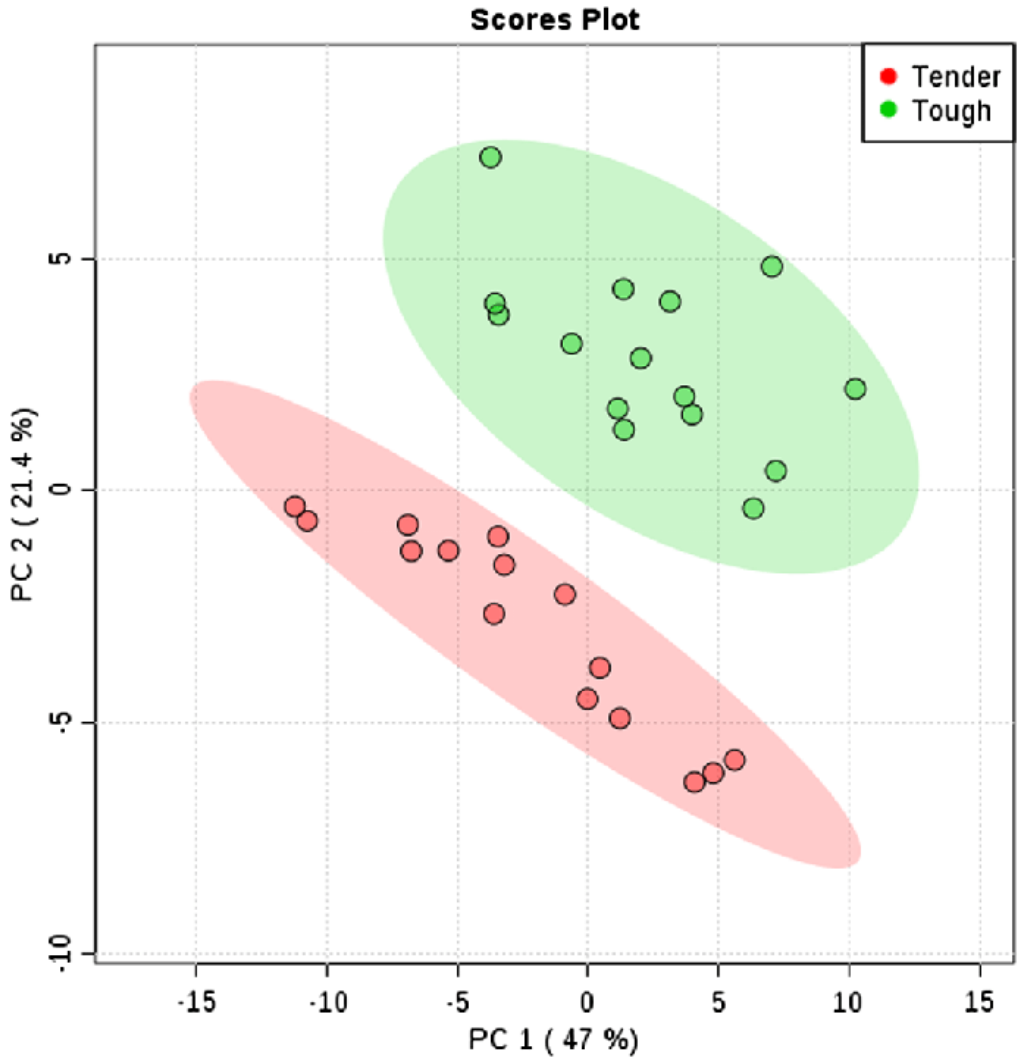
   Component directions are perpendicular to one another.

3. **Maximum variance in the first component**

   PC1 captures the greatest variance; later components capture the greatest remaining variance subject to orthogonality.

4. **No outcome information is used**

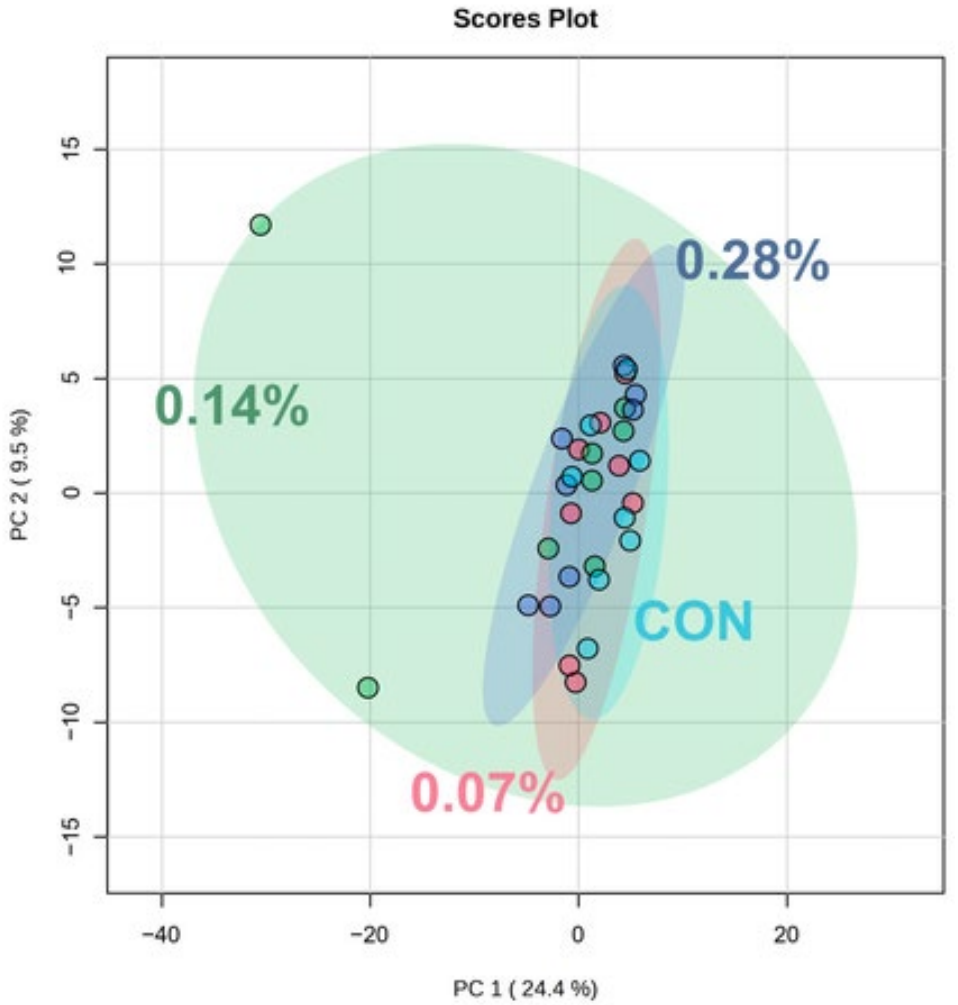
   Component directions are learned from the input features —not a target or outcome.



Principal Component Analysis (PCA)

Scores: positions of observations Each observation has a coordinate on each component. A score plot places observations on these new axes.

Explained variance: The percentage of total input variation captured by each component—not prediction accuracy.

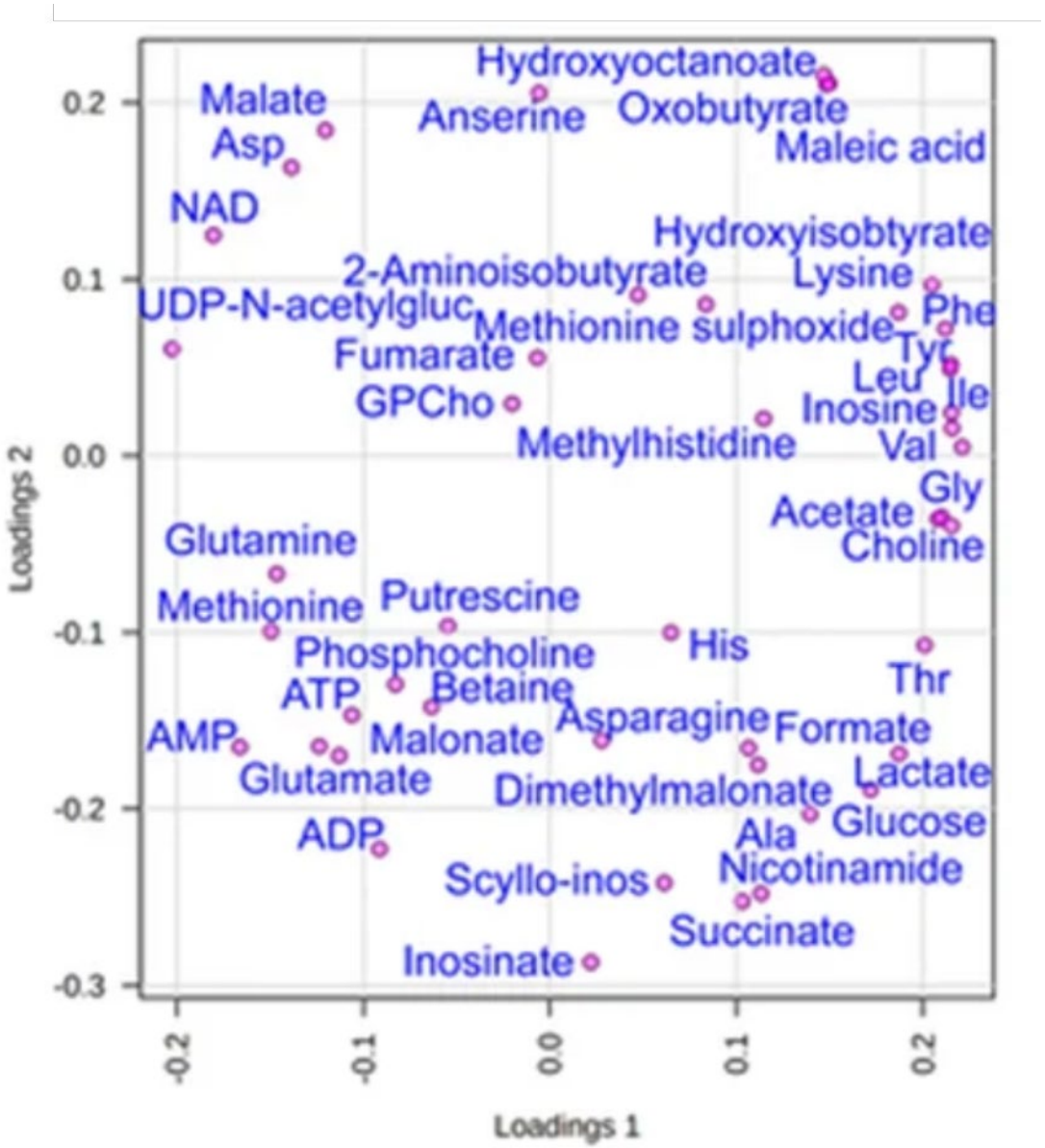


(B)

Principal Component Analysis (PCA)

Loadings: contribution of variables

- Signed weights describe how variables form a component.
- Larger absolute weights indicate greater influence on that component.



Partial Least Squares (PLS)-Supervised

Build components that connect predictor information to an outcome.

1 Components constructed using predictors and outcome

Predictor weights are learned using both X and y, seeking components that covary with the outcome.

2 Useful with correlated predictors

Summarizes overlapping predictor information in a smaller set of components.

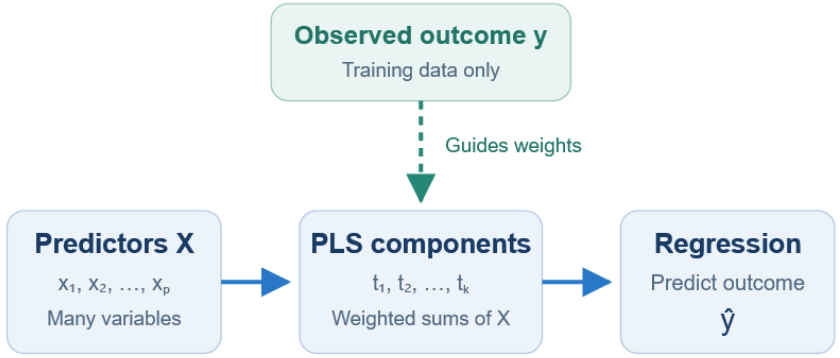
3 Common in chemometrics and omics

Examples include spectral measurements, gene expression, and metabolomic profiles.

4 Supervised dimensionality reduction

Unlike PCA, the outcome helps determine which predictor directions are retained.

The outcome guides learning



New observations: use X and the learned model—not y.

Conceptual workflow for PLS regression. The outcome informs component learning; component scores are computed from predictors.

PCA

Finds directions of greatest variation in X.

Outcome is not used.

PLS

Finds predictor components that covary with y.

Outcome guides learning.

Key distinction: PCA summarizes predictor variation; PLS uses the outcome to guide that summary.

Partial Least Squares Discriminant Analysis (PLS-DA)

Partial Least Squares Discriminant Analysis: use class labels to guide component learning.

1 PLS adapted for classification

The outcome represents class membership, often through indicator-coded labels.

2 Latent components separate outcome groups

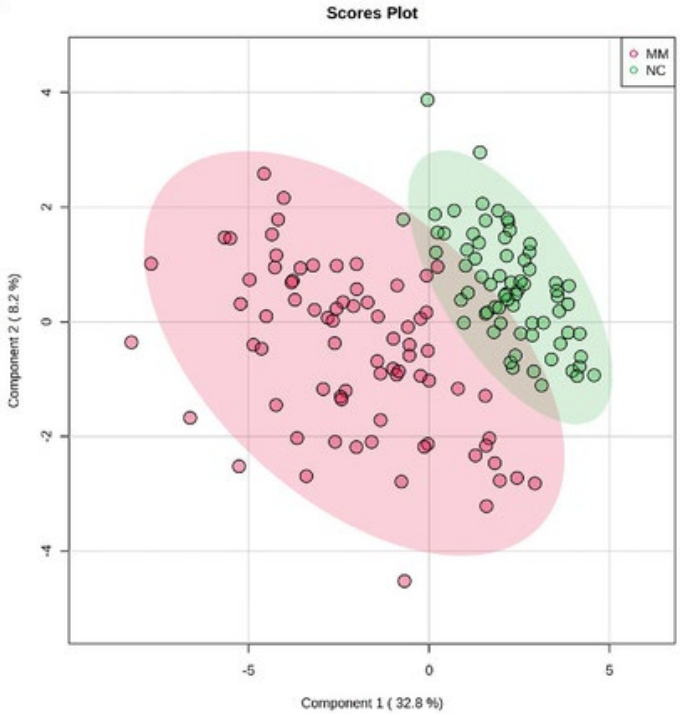
Components are learned to capture predictor patterns associated with the known classes. Separation is not guaranteed.

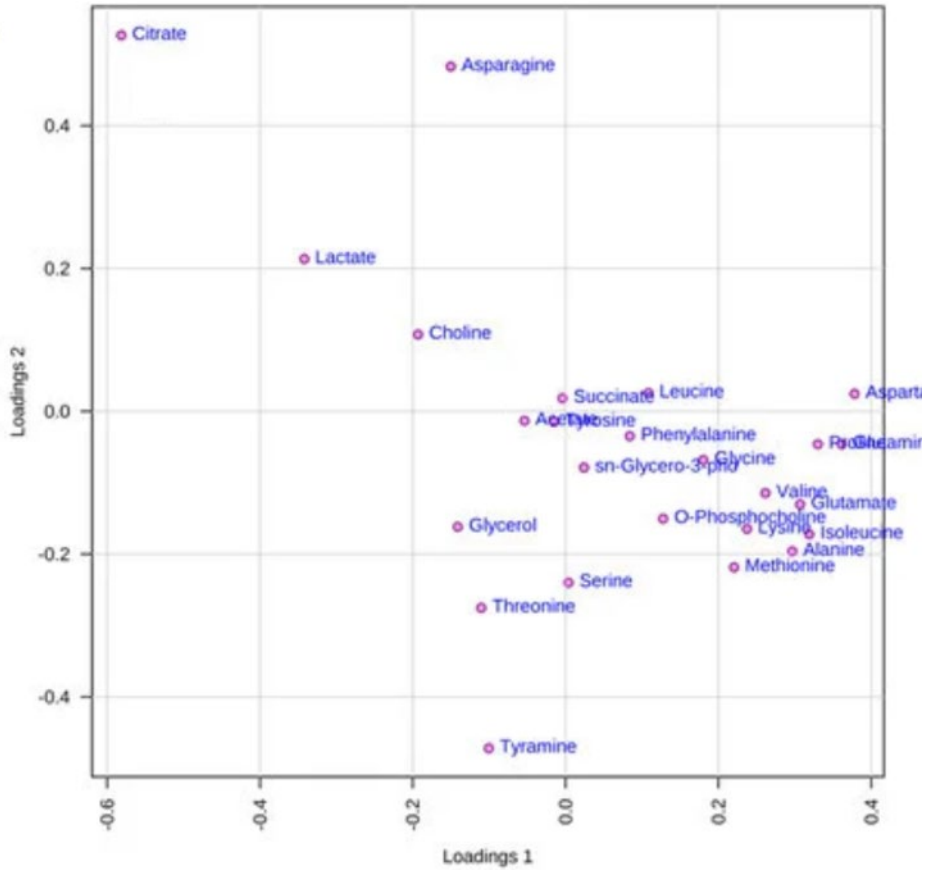
3 Scores and loading plots

Scores show observations; loadings describe variable contributions to the components.

4 Requires rigorous validation

Evaluate predictions for unseen observations, not just separation in the training plot.





Read the plots—but judge the model by out-of-sample predictions.

PLS-DA: Common Pitfalls

Strong-looking separation can coexist with poor prediction.

1 Apparent separation can occur by chance

Because labels guide fitting, even arbitrary groups can appear separated.

2 Overfitting in small, high-dimensional datasets

Many variables and few observations make noise easier to fit.

3 Select components within cross-validation

Use inner folds for tuning and outer folds for performance estimation.

4 Use permutation testing and external validation

Compare against shuffled labels, then test the locked model on independent data.

A leakage-free validation workflow

1. Set aside an outer test fold

Keep it out of model development.

Inside the remaining data: use inner cross-validation to select components. Fit scaling and feature selection within each training fold.

↓

2. Refit and predict the outer test fold

Repeat across outer folds; summarize held-out classification performance.

Permutation test

Shuffle labels appropriately for the study design. Repeat the full tuning and validation workflow.

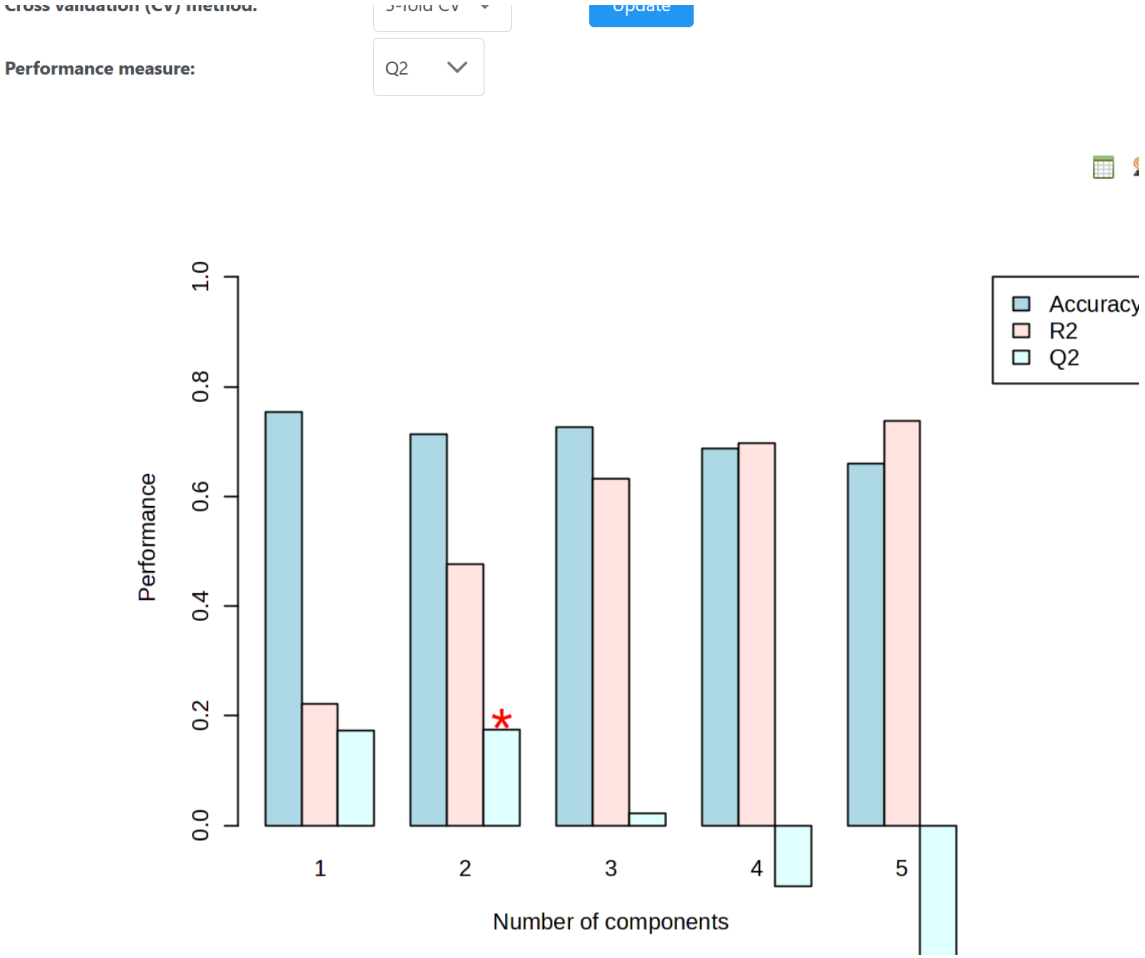
External validation

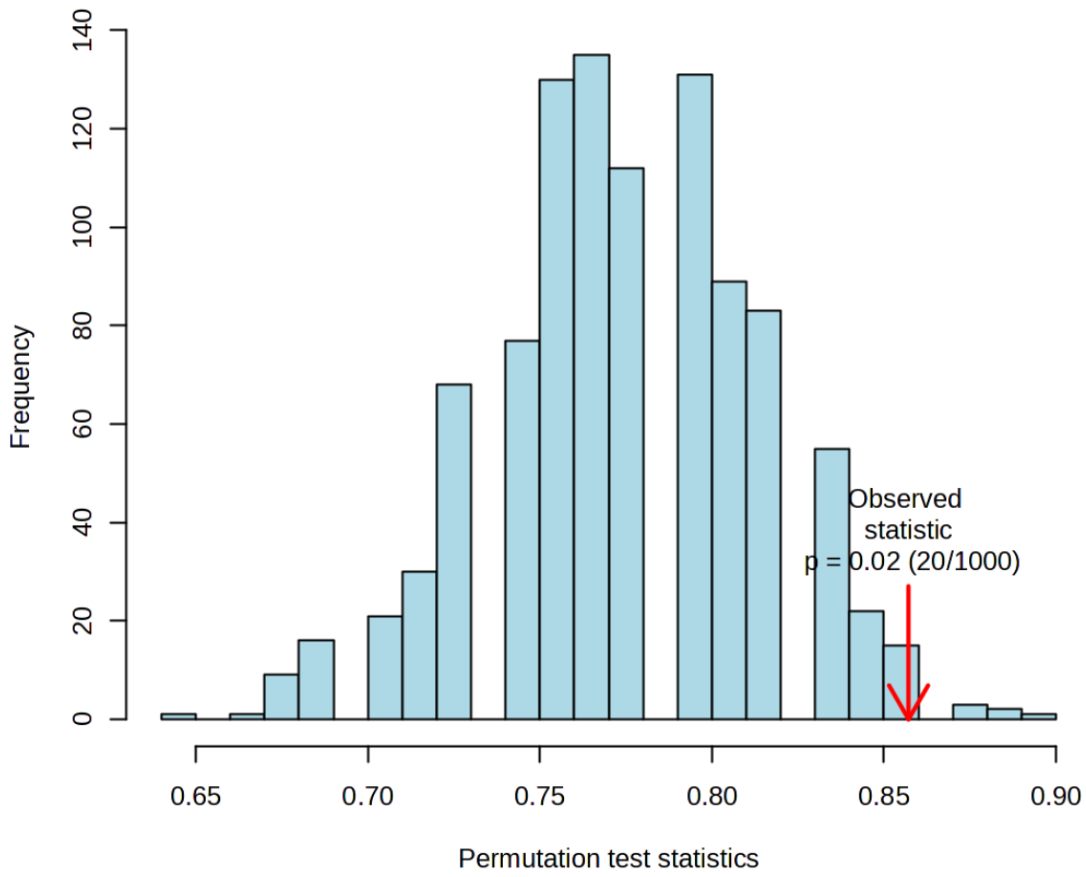
Apply the locked pipeline to an independent dataset. Do not retune on that dataset.

Avoid leakage: Do not choose variables or components using the observations reserved for testing.

Bottom line: Validate the entire modeling pipeline—not just the final fit.

Partial Least Squares Discriminant Analysis (PLS-DA)





https://www.metaboanalyst.ca

Clustering: How Do We Choose a Clustering Method?

Start with the data and the scientific question—not the most complicated algorithm.

Your data or goal	Method to consider	Why?
Numeric features; compact, roughly round groups	**K-means**	Simple centroid-based grouping; fast baseline.
Continuous measurements; overlapping profiles; membership uncertainty matters	**Latent profile analysis (LPA)**	A statistical mixture model gives probabilities of belonging to each profile.
Categorical indicators, such as yes/no responses	**Latent class analysis (LCA)**	Models response probabilities within latent classes.
Explore nested groups at different levels	**Hierarchical clustering**	A dendrogram displays a hierarchy of clusters.
Irregular shapes; some observations may be noise	**DBSCAN / HDBSCAN**	Find dense regions without requiring a fixed number of clusters in advance.

First define "similar"

Choose relevant variables and a defensible distance or probability model. For distance-based methods, consider scaling when units differ.

PCA is not a clustering method

PCA creates a lower-dimensional representation. Clustering may follow, but retained components must preserve information relevant to grouping.

Key message: Different methods define a "cluster" differently. There is no universally best clustering algorithm.

Clustering Mixed Clinical Data

Continuous and Categorical Features

Example: Age, BMI, and blood pressure + diabetes status, medication category, and symptom severity.

Method	Main idea
Gower Distance + K-Medoids (PAM)	Combines mixed-variable differences and groups patients around representative patients.
Gower Distance + Hierarchical Clustering	Builds a hierarchy of patient groups, displayed as a dendrogram.
K-Prototypes	Combines numerical means and categorical modes to define cluster centers.
Mixed-Indicator Mixture Models	Models continuous and categorical features together and estimates membership probabilities.

Remember: Gower is a distance measure—not a clustering algorithm. Specify variable types correctly before clustering.

Clustering

Patient	Age (years)	Resting BP (mmHg)	Serum cholesterol (mg/dL)	Chest pain type	Exercise-induced ang
P01	45	120	190	Non-anginal pain	No
P02	48	125	205	Non-anginal pain	No
P03	65	150	270	Typical angina	Yes
P04	68	155	280	Typical angina	Yes
P05	56	135	230	Atypical angina	No

Clustering

Can we identify groups of clinically similar patients and select an actual patient to represent each group?

A. K-means using numerical category codes

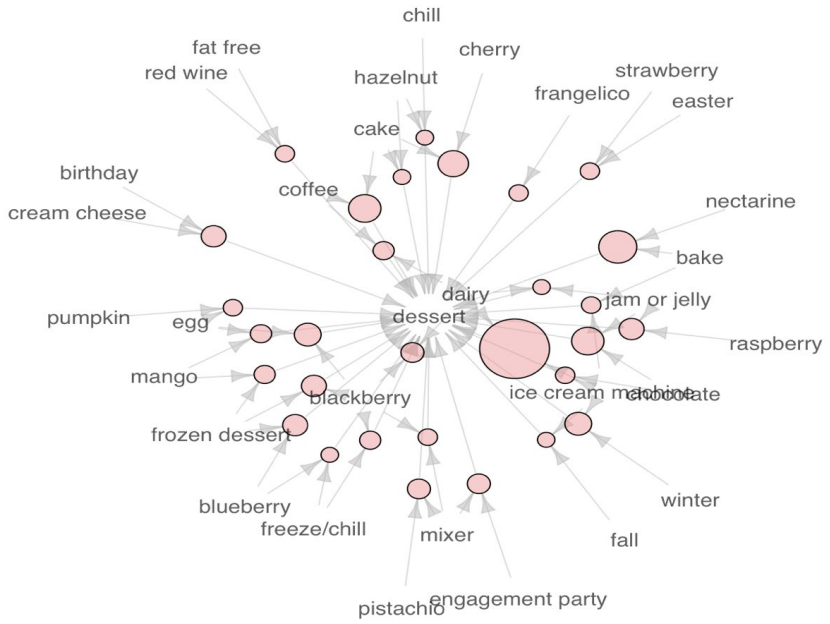
B. Gower distance + K-medoids (PAM)

C. K-prototypes

D. Gower distance + hierarchical clustering alone

Clustering

Gower Measures Similarity; PAM Selects Representatives



1. Gower Distance

"How different are these two patients?"

Gower combines differences across **numerical and categorical features** into one dissimilarity score.

Feature type	Comparison
Numerical Age, blood pressure, cholesterol	Absolute difference divided by the feature's range in the dataset
Nominal / symmetric binary Chest pain type, angina status	Same category = 0 Different category = 1

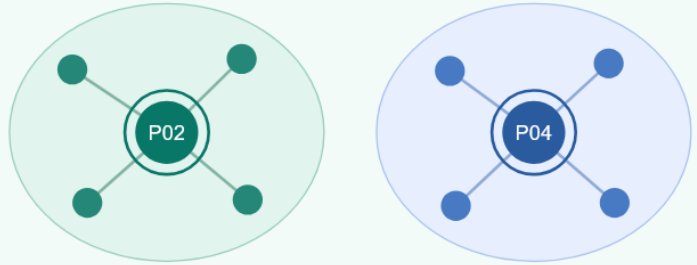
Gower distance
= weighted average of feature-level differences

Example: P01 versus P02
Age: 45 versus 48 → a small numerical difference.
Chest pain: both non-anginal → difference = 0.
Angina: both "No" → difference = 0.

Lower distance → more similar patients.
Gower creates a patient-to-patient distance matrix. It does **not** create clusters by itself.

2. PAM: Partitioning Around Medoids

"Which real patients represent the groups?"



Actual patient representative

Actual patient representative

A medoid is an actual observed patient chosen as a central representative using the distances. It is not a synthetic "average patient."

Given a chosen number of clusters, PAM searches for medoids that reduce the total distance from patients to their nearest medoid.

With Gower + PAM:
Compare mixed clinical features → form groups → identify actual patient representatives.

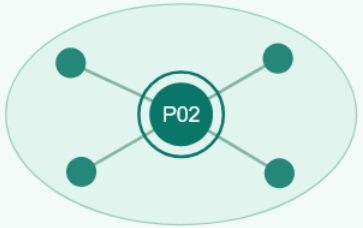
Gower: defines patient dissimilarity → **PAM:** clusters patients around real representatives

Why Gower + PAM Rather Than the Alternatives?

The key requirement is not just to form groups: each group must have an actual patient as its representative.

Gower + PAM

Directly meets both requirements



Representative = an observed patient

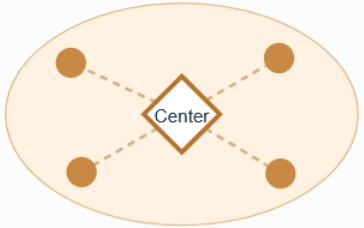
Mixed features: Gower handles numerical and categorical differences.

Representative: PAM selects an actual patient as a medoid.

Example representative: P02
Age: 48 years
BP: 125 mmHg
Cholesterol: 205 mg/dL
Chest pain: non-anginal
Exercise-induced angina: No

K-prototypes

Handles mixed data; center may not be a patient



Diamond = a synthetic prototype

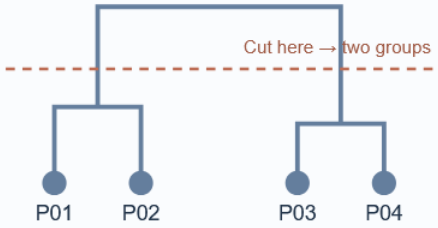
Mixed features: supported by the standard K-prototypes method.

Representative: numerical means + categorical modes—not necessarily an actual patient.

If P01 and P02 form a cluster:
Mean age: 46.5 years
Mean BP: 122.5 mmHg
Mean cholesterol: 197.5 mg/dL
Shared categories become the modes.
This center is neither P01 nor P02.

Hierarchical Clustering

Forms groups; representative selection is extra



Mixed features: can be handled using Gower distance with suitable linkage.

Representative: the dendrogram does not automatically identify a medoid.

A valid alternative workflow:
Gower distance
→ hierarchical clustering
→ cut the tree into groups
→ select a medoid within each group.
Possible, but requires an extra step.

Best direct match: Gower + PAM
Mixed clinical features + actual patient representatives

Clustering

Can we identify patient profiles and estimate each patient's probability of belonging to each profile, allowing uncertain classification?

A. Gower distance + K-medoids (PAM)

B. K-prototypes

C. Mixed-indicator latent class / finite mixture model

D. Continuous-only latent profile analysis

Machine Learning Is Not Magic

1. Machine learning is **not magic.**
2. It is systematic **function estimation from data.**
3. **Data quality and study design** often matter more than model novelty.

Take Away Messages

Complexity Brings Both Power and Risk

1. Complex models can discover **complex patterns.**
2. Complexity creates **additional risks.**
3. **Validation, transparency, and monitoring** are essential.

Take Away Messages

Prediction Is Only Part of Clinical Decision-Making

1. Prediction is **only one part** of healthcare decision-making.
2. **Clinical expertise, patient values, and causal evidence** remain necessary.
3. AI should **augment—not replace—responsible judgment.**

Take Away Messages

Comment | Open access | Published: 28 February 2024

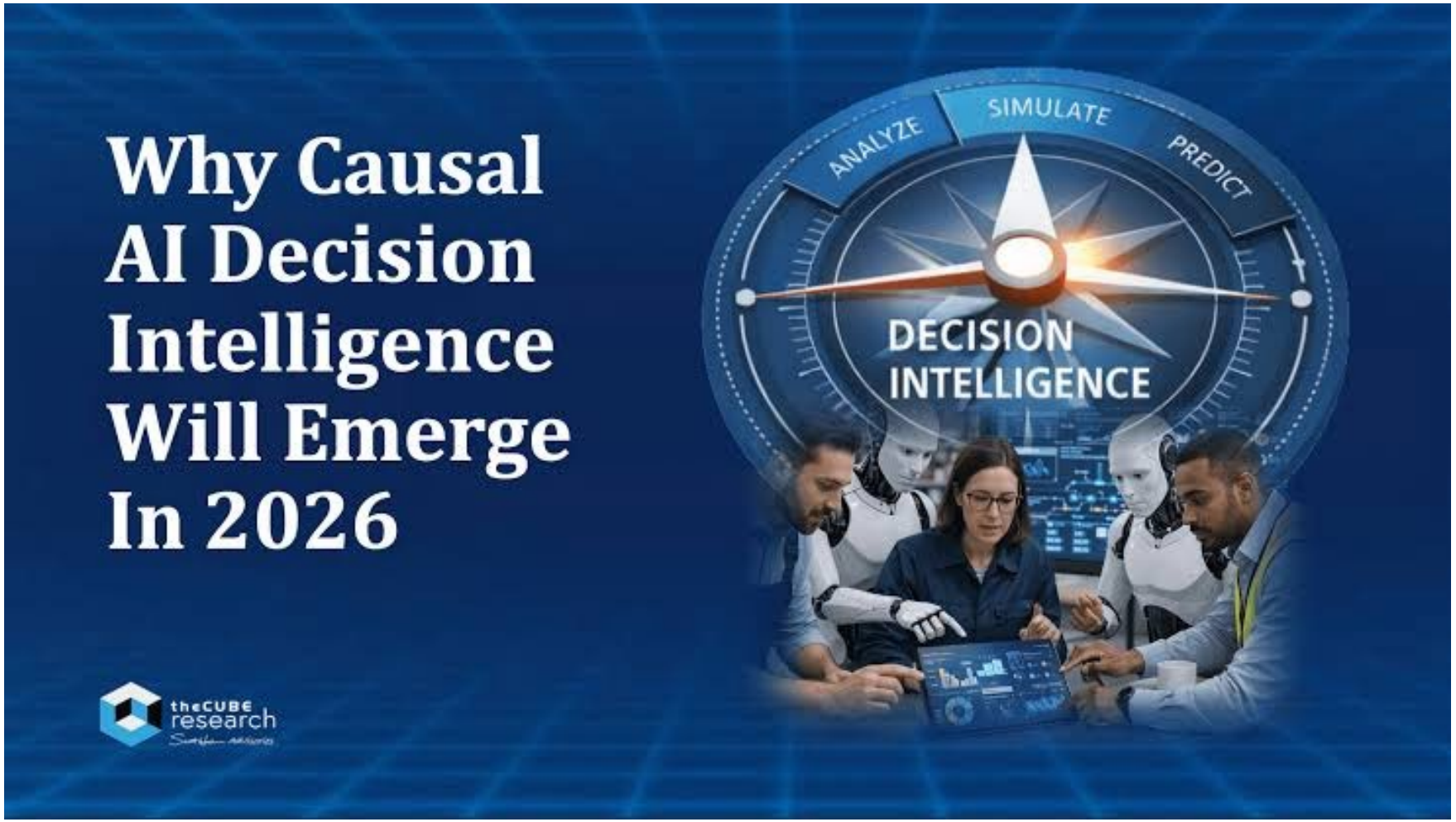
All models are wrong and yours are useless: making clinical prediction models impactful for patients

Florian Markowetz

npj Precision Oncology **8**, Article number: 54 (2024) | Cite this article

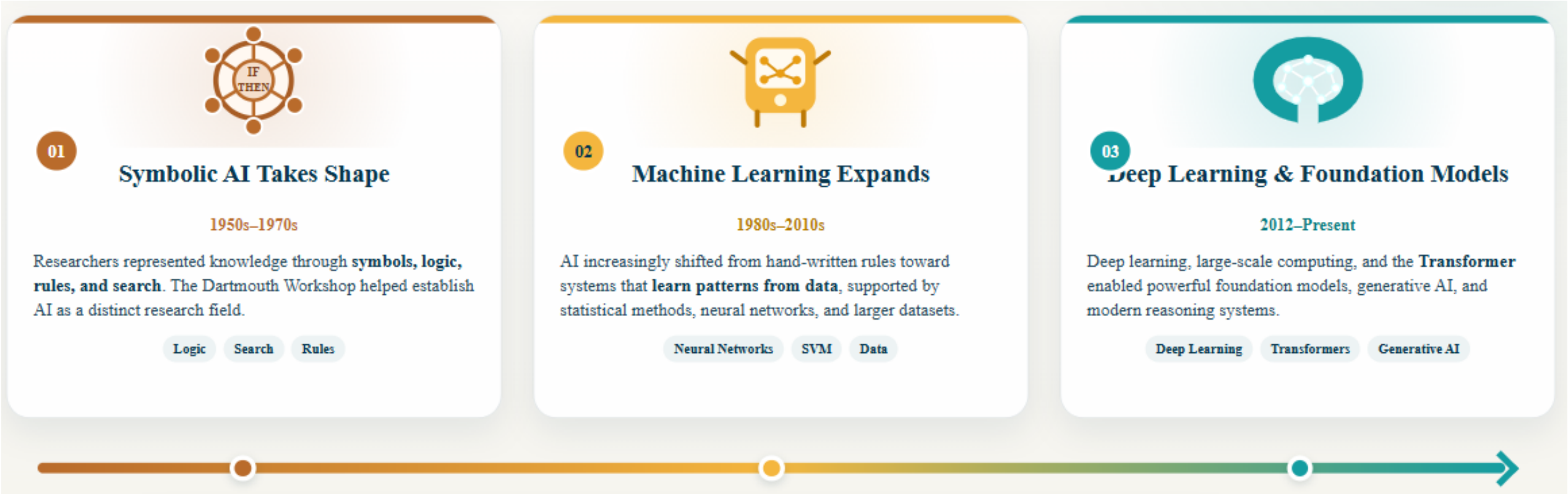
Most published clinical prediction models are never used in clinical practice and there is a huge gap between academic research and clinical implementation.

Take Away Messages



From Turing's Question to Modern AI

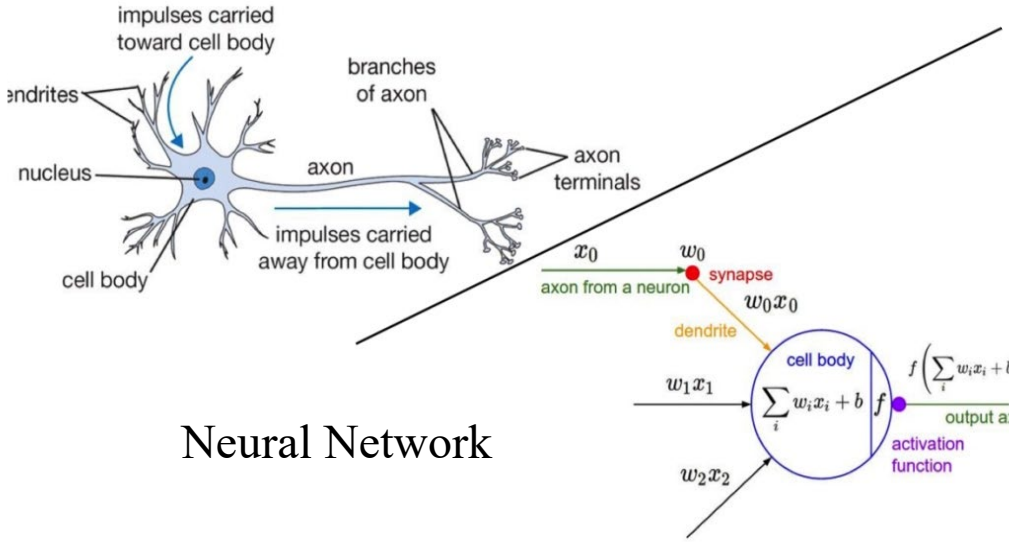
A brief roadmap of how artificial intelligence has evolved



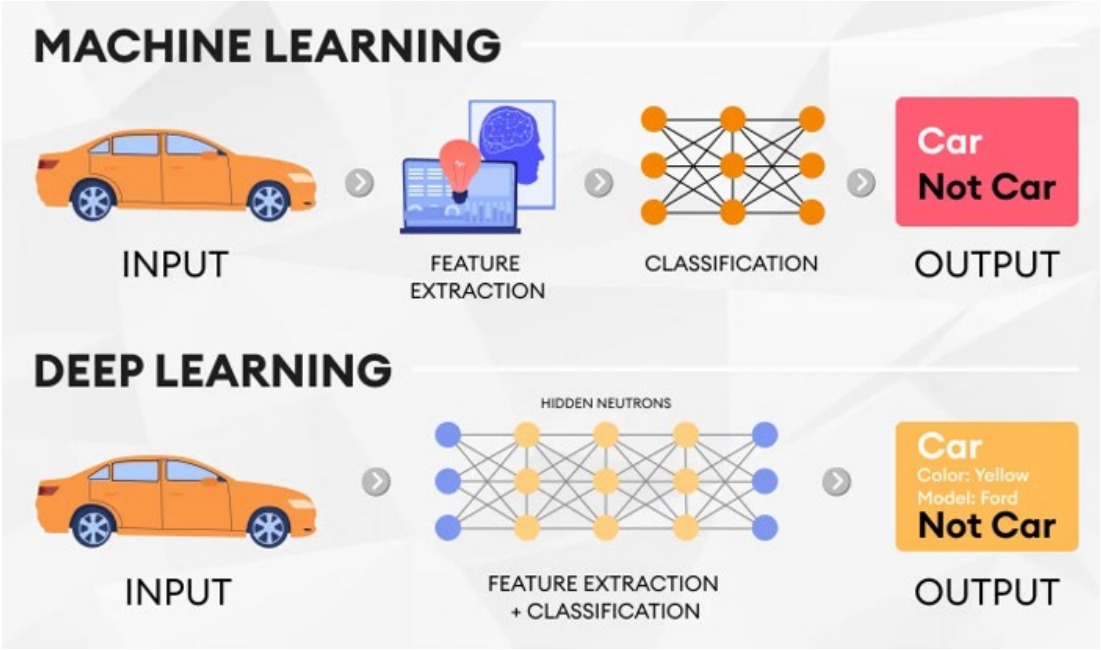
The methods changed—from rules to learning to foundation models — but the goal remained the creation of intelligent behavior.

Deep Learning: Learning Features from Data

Deep learning is a type of machine learning that uses neural networks with multiple layers to learn patterns and representations from data.



Neural Network



The core philosophy of connectionism: Connectionism, also known as the bionics school or physiological school, **operates on the primary principle of constructing neural networks to simulate the ways human brain neurons connect and process information.**

The key idea: automatic feature learning

Symbolic AI----Machine learning----Neural Network

Depth Became Trainable Only After Three Tricks

After 2006, "deeper" finally stopped meaning "worse" — and by 2020 every modality was using the same block.

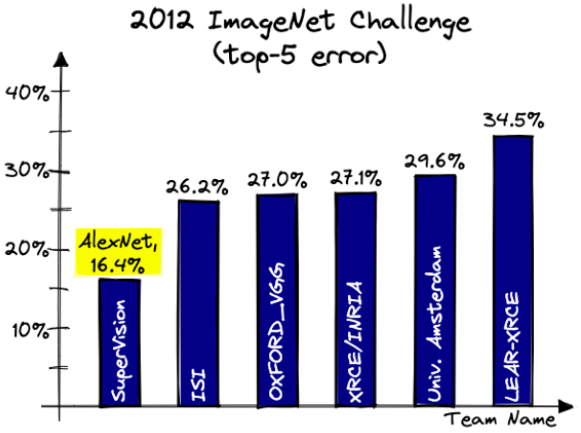
images	→	text	→	signals	→	structure	→	one block
CT · retina		clinic notes		accelerometry		protein · graph		the same Transformer

2006	2009	2012	2015	2017	2018–20	2023–26
Layer-wise pretraining	**ImageNet**	**AlexNet**	**ResNet · BatchNorm**	**Transformer**	**BERT · GPT · ViT · DDPM**	**Foundation · reasoning · SSM**
depth becomes optimisable	14M labelled images	error drops 26% → 16%	152 layers, trainable	recurrence removed	pretraining takes every modality	multimodal; linear-cost sequences



Fei-Fei Li

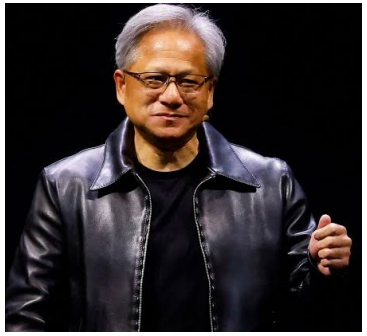




Graphics processing unit (GPU)



The term was coined by NVIDIA as well as CUDA



Jensen Huang

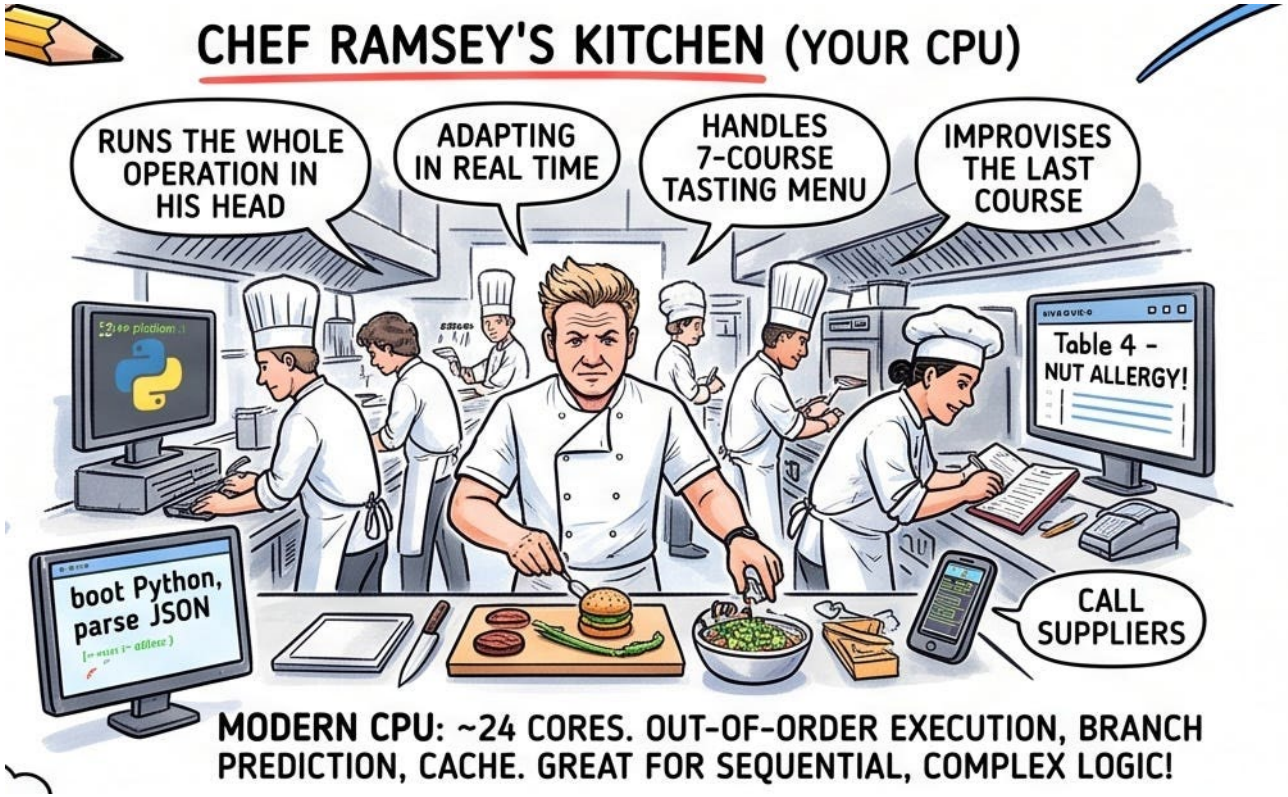
The world's most valuable company.

澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

Krizhevsky, Sutskever & Hinton, NIPS 2012 · He et al., CVPR 770–778, 2016 · Vaswani et al., NIPS 2017

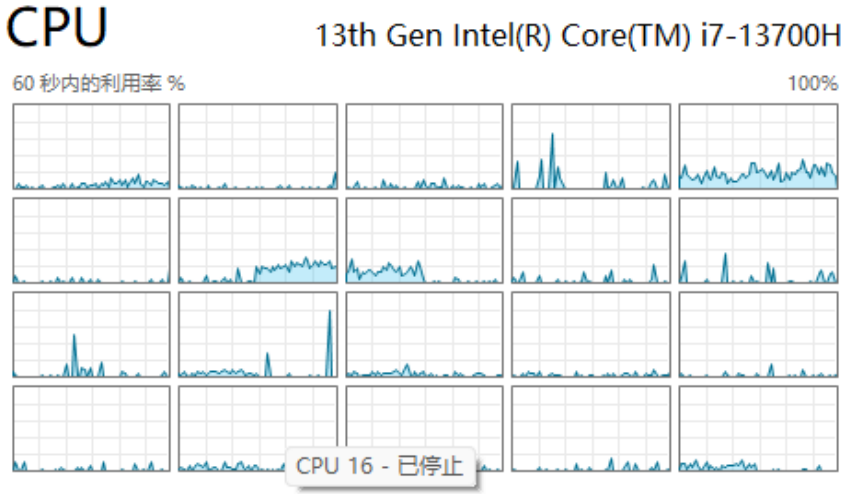
Graphics processing unit (GPU) vs CPU (Central processing unit)

CPU (Central processing unit)



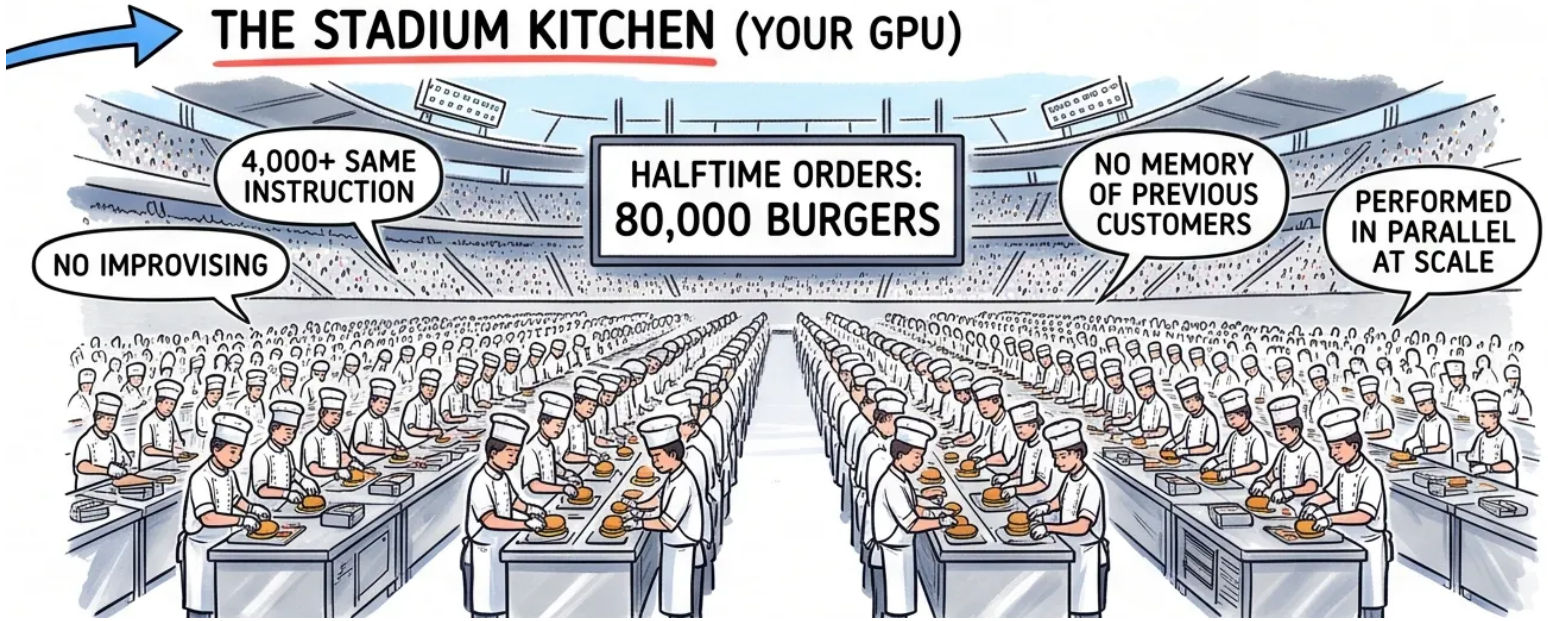
- A modern high-end CPU has maybe 24 cores. Each one is capable. Out-of-order execution, branch prediction, multiple levels of cache, clock speeds above 5 GHz.
- Ramsey's kitchen can handle a reservation change mid-service, run the books afterward, and call suppliers in the morning.
- When you boot Python, parse a JSON file, or run a Flask server, Ramsey's kitchen is exactly what you want

Ctrl + Shift + Esc---Performance--CPU-
Change graph to---Logical processors



Graphics processing unit (GPU): SIMD/SIMT

SIMD （**Single Instruction, Multiple Data**): Massive parallel processing

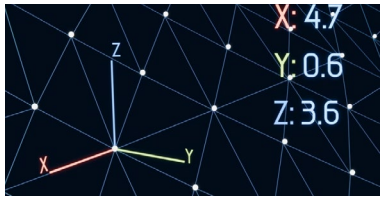


The head chef picks up the mic: “Everyone, cut onions (SIMD)



The cowboy hat is built by 11480 vertices. Each with X, Y, Z coordinates.

A 2048 CUDA core can handle 2048 vertices at a time.



While CPU can only do 24/32 at once (24-core or 32-core)

A neural network isn’t one complex sequential task. It is a million of simple repetitive calculations; mostly matrix calculations that can all be performed in parallel. This is a perfect design for deep learning

Graphics Card

Graphics Card vs GPU





Semiconductors (GPU Computing Power)

Global Semiconductor Landscape: Challenges of Decoupling



United States

- EDA/IP (74%)
- Logic devices (67%)
- Manufacturing equipment (41%)



China

- Packaging and testing (38%)
- Wafer fabrication (16%)
- Raw materials (13%)



EU

- EDA/IP (20%)
- Manufacturing equipment (18%)

No single region controls the entire semiconductor supply chain.

EDA = Electronic Design Automation; IP = Intellectual Property.

What Actually Changed Was Scale

Six numbers explain the last fifteen years better than any architecture diagram （2012 to now）

ImageNet top-5 error

26% → 3%

Network depth

8 → 152

layers

Training compute

~10^8 × Ten to the eighth

Labelled images available

14 M

ImageNet

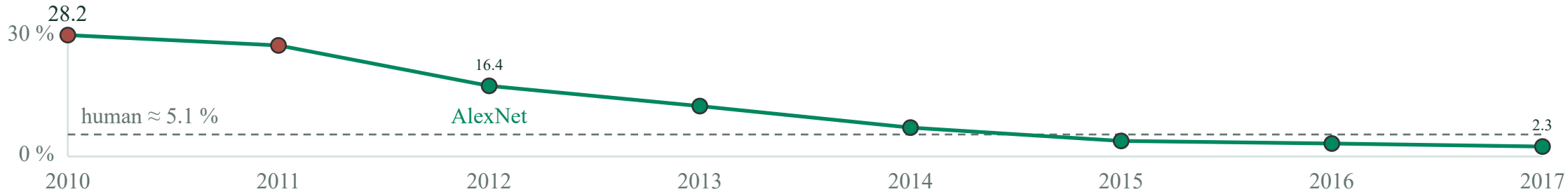
Parameters

60 M → 10^{12}

Sixty million to ten to the twelfth

Time to reproduce a 2012 result

6 days → minutes



- Your bottleneck is data, not GPUs.

ILSVRC official results; published compute-trend estimates. Order-of-magnitude teaching figures.

What is an Artificial Neural Network?

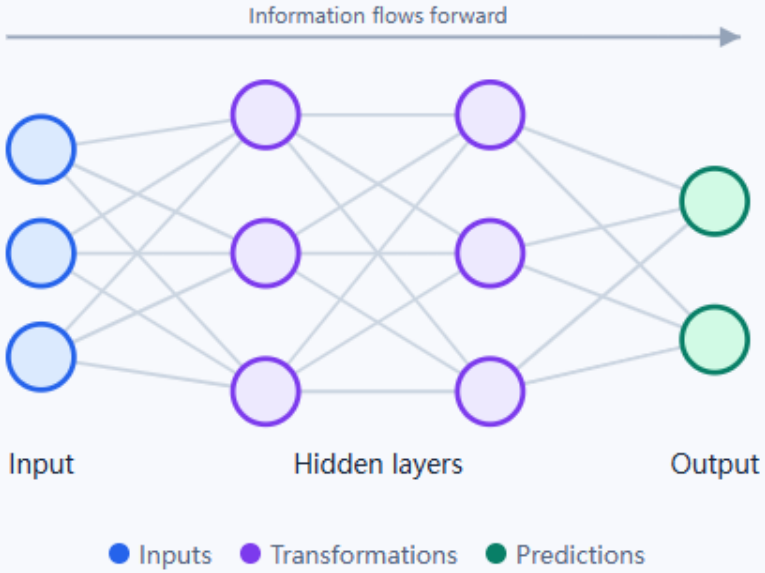
Neuron / Node | Layer | Deep Learning

Everyday analogy: A committee reviews information, discusses it in stages, and gives an answer.

A trainable system of connected mathematical units

- **Input layer — receives the information.**
  Examples: image pixels or a patient's blood pressure.
- **Hidden layers — transform the information.**
  Neurons combine signals to build useful features.
- **Output layer — produces a prediction.**
  Example: "straw hat" versus "not a straw hat."

A **neuron** is a small computation unit. A **layer** is a group of these units.



Why "deep" learning? Deep neural networks use multiple layers of learned transformations. More depth does not automatically mean a better model.

Inspired by biological nervous systems, but not a literal simulation of a human brain.

Weights and Bias: What Matters Most?

Input x | Weight w | Bias b | Score z

A neuron combines inputs into a score before applying an activation.

Everyday analogy: "Should I go camping?" Weather and company influence the decision.

- **x_1 = good weather:** 1 for yes; 0 for no.
- **x_2 = a friend is joining:** 1 for yes; 0 for no.
- **$w_1 = 10$, $w_2 = 1$:** In this example, weather matters much more than company.
- **b = −5:** A stay-at-home tendency shifts the score downward.

$$z = x_1 w_1 + x_2 w_2 + b$$

A **weight** scales an input's contribution. A **bias** shifts the score, even when inputs are zero.

Bias (b): like your character

Try the camping decision

Keep the weights and bias fixed. Change the inputs:

Good weather, x_1 — Yes: 1

Friend joining, x_2 — No: 0

$$z = (1 \times 10) + (0 \times 1) - 5 = 5$$

Sigmoid output: **0.9933**
Toy decision: **Go camping**

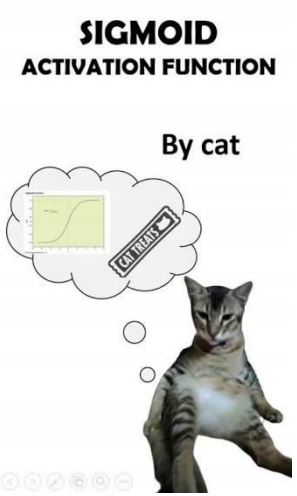
We use a sigmoid activation and choose "go" when its output is at least 0.5. This is a teaching example, not a trained model.

Learnable parameters: During training, the model adjusts weights and biases. They are not usually hand-picked like the values above.

Activation Functions: More Than an On/Off Switch

An activation converts a neuron's score into its output signal (**on/off switch**).

Three main activation functions: ReLU (Rectified Linear Unit) , Sigmoid and Softmax



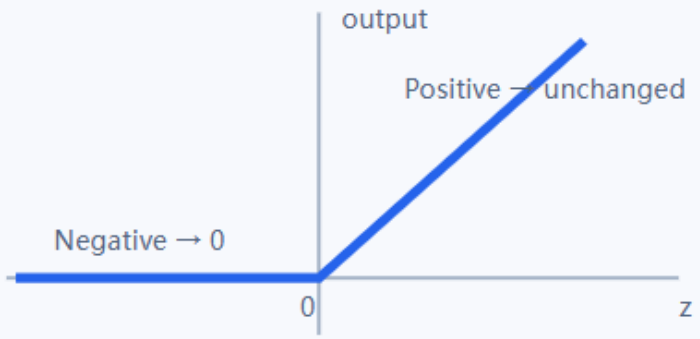
Sigmoid turns an unreadable score (5.0) into a clear probability (99.33% Cat).

Activation Function | ReLU | Sigmoid | Nonlinearity

Everyday analogy: A signal gate or dimmer: it changes how much information passes through.

ReLU: keep positive values

$$\mathrm{ReLU}(z) = \max(0, z)$$



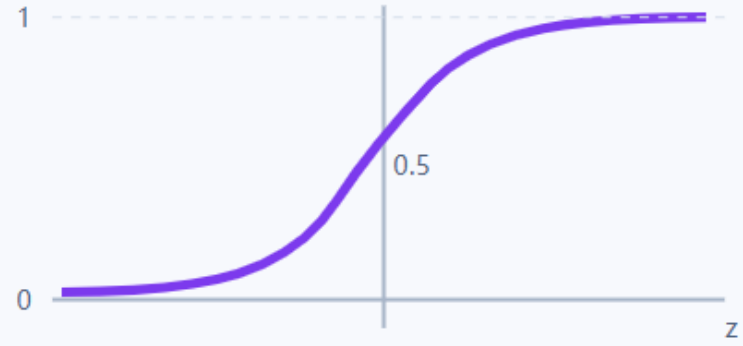
Examples: −3 → 0; 0 → 0; 8.5 → 8.5.

Often used in hidden layers.

Softmax

Sigmoid: squash to 0–1

$$\sigma(z) = 1 / (1 + e^{-z})$$



Examples: −5 → 0.0067; 0 → 0.5; 5 → 0.9933.

Can be used for a binary-classification output.

Why activations matter: Nonlinear activations let networks learn patterns that stacked linear transformations alone cannot express.

Activation Functions: More Than an On/Off Switch

Softmax: Used for Multi-class Classification (Choose one from many, e.g., Cat vs. Dog vs. Bird)

Turn messy scores into beautiful probabilities!

What does Softmax do?

It converts a group of raw class scores (**logits**) into a probability distribution.

- Each class receives a probability between 0 and 1.
- The probabilities add up to **100%**.
- A higher score receives a higher probability.

$$p_i = e^{z_i} / \sum_{j=1}^{K} e^{z_j}$$

zi = score for class i; K = number of classes. Each probability depends on all the class scores.

Analogy: Divide one probability pie among competing answers.

Example: Classify one image

Scores → Softmax → Probabilities

Class	Raw score	Probability
Cat	2.0	66.5%
Dog	1.0	24.5%
Bird	0.0	9.0%

Choose the highest probability → **Cat**

Softmax produces probabilities. Choosing the largest one is a separate decision rule.

Remember: Softmax → one of several mutually exclusive classes. **If multiple clinical findings can coexist, use separate sigmoid outputs instead.**

Why Do We Need ReLU or Other Activation Functions?

More layers alone are not enough. Nonlinear activations make depth more expressive.

1. Without nonlinear activations

Consider two layers of weighted sums plus biases:

$$h = W_1x + b_1$$
$$z = W_2h + b_2$$

Substitute the first layer into the second:

$$z = W_2(W_1x + b_1) + b_2$$
$$= (W_2W_1)x + (W_2b_1 + b_2)$$
$$= \mathbf{W_{new}x + b_{new}}$$

The combined weights and biases are simply **another set of weights and a bias.**

Even 1,000 affine layers are equivalent to one affine layer.

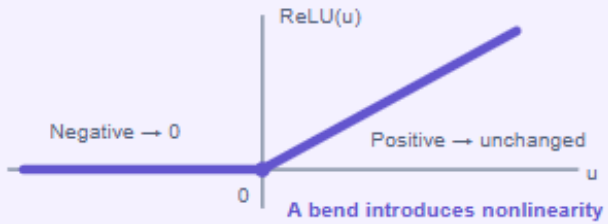
Affine = a linear transformation plus a bias. Adding depth alone does not add nonlinear expressiveness.

2. With a nonlinear activation

Insert ReLU between the layers:

$$h = \mathbf{ReLU}\,(W_1x + b_1)$$
$$z = W_2h + b_2$$

ReLU(u) = max(0, u): negative values become zero; positive values pass through.



The network can now learn nonlinear relationships, rather than only a single affine mapping.

With ReLU between layers, the network generally cannot be rewritten as one affine layer. Other nonlinear activations include sigmoid and tanh.

Without nonlinearity: more layers ≈ one affine layer.
With nonlinearity: depth can build richer representations.

Forward Propagation: The First Attempt

Data travels from the input through the network to a prediction.

Forward Propagation | Prediction | Parameter Initialization

Everyday analogy: A student attempts an exam before receiving feedback.

Input
Image pixels

→

Hidden layer
Weights + bias
then activation

→

Next layer
Repeat transformations

→

Output
Prediction

Weights initialized randomly
Biases: initialized to zero

Action function (Softmax)

Example input: a cat



An untrained model might say...

Cat	10%
Tank	80%
Other	10%

Confident-looking does not mean correct. These invented scores illustrate a possible bad prediction.

Weights are often initialized randomly; biases are often initialized to zero. Before training, predictions need not be meaningful.

Forward propagation produces the prediction. By itself, it does not change the model's weights or biases.

Loss Functions: A Mathematical Grading Rule

Target / Label | Loss Function | Cross-Entropy | Training Objective

Everyday analogy: A teacher uses a marking rule—not just "right" or "wrong."

Grade the homework

Compare the prediction with the target and assign a penalty.

- **Target:** The correct class is "cat."
- **Prediction:** The model assigns a probability to each class.
- **Loss:** A number measuring how poorly the prediction matches the target.

For this classification example, use **cross-entropy for the correct class**:

$$\text{Loss} = -\ln(p_{\text{correct class}})$$

A low probability for the correct answer gives a large penalty.

An untrained model might say...



Confident-looking does not mean correct. These invented scores illustrate a possible bad prediction.

Poor prediction, higher loss

Same cat, different predictions

P(cat)	Loss	Meaning
0.10	2.303	Large penalty
0.50	0.693	Smaller penalty
0.90	0.105	Small penalty

Teacher punish

Other tasks use other losses. For predicting a numerical value, squared error is one option:

$$\text{Loss} = (\text{prediction} - \text{target})^2$$

Training aims to reduce an objective based on loss. Zero loss is not always achievable or desirable; performance on unseen data matters too.

Backpropagation and Gradient Descent

- First calculate how to change parameters (**Backpropagation**); then make the changes (**Gradient descent**).

- Once we have the loss, we cannot just tell the AI to 'get smarter.' We must tell each neuron exactly how to adjust its weights—whether to turn them up or down

Backpropagation | Gradient | Gradient Descent | Learning Rate η

Everyday analogy: Review an incorrect answer, identify which decisions contributed, and adjust the decision rules.

1. Backpropagation: calculate gradients

Work backward through the computation using the **chain rule**.

A gradient describes how the loss would change with a small change in a parameter.

2. Gradient descent: update parameters

Move a small step in the direction opposite the gradient.

$$w \leftarrow w - \eta \cdot \partial Loss/\partial w$$
$$b \leftarrow b - \eta \cdot \partial Loss/\partial b$$

→ **η, the learning rate,** controls the step size. Steps that are too large may overshoot; very small steps can be slow.

Optimizer

Imagine walking downhill



Imagine you are blindfolded on a mountain. The gradient tells you which way is down, and the learning rate decides how big your steps are. If it's too big, you overshoot; if it's too small, it takes forever!"

Important distinction: Backpropagation computes gradients. An optimizer, such as gradient descent, uses them to update parameters.

The Training Loop: Predict, Measure, Improve

Repeat these stages using training examples.

Training Loop | Batch | Epoch | Generalization

1

Forward

Feed in a batch of inputs.

Apply weights, biases, and activations.

Result: predictions.

2

Loss

Compare predictions with the targets.

Compute the training objective.

Result: a penalty to minimize.

3

Backward + Update

Backpropagation computes gradients.

The optimizer updates weights and biases.

Result: adjusted parameters.

↻ Repeat with the next batch, then continue over more epochs.

Can You Explain How a Neural Network Learns?

Use the vocabulary, then test your understanding.

Quick vocabulary review

Neuron:
A computation unit.

Layer:
A group of units.

Weight:
Scales an input contribution.

Bias:
Shifts a neuron's score.

Activation:
Transforms the score.

Forward pass:
Produces a prediction.

Loss:
Measures prediction mismatch.

Gradient:
Local sensitivity of loss.

Backpropagation:
Computes gradients backward.

Optimizer:
Updates parameters.

Learning rate:
Controls update size.

Epoch:
One pass through training data.

- **One-sentence summary**: A neural network learns by making predictions, measuring their errors, and adjusting its weights and biases using gradients.

Quiz 1

What does ReLU do with −4 and +7?

- ReLU(−4) =
- ReLU(+7) =

2. Does a forward pass update the weights?

Neural Network Architectures

Different Data, Different "Superpowers"
An architecture defines how a neural network organizes computation.

CNN

Convolutional
Neural Network

NATURAL FIT

Images and spatial grids

CORE MECHANISM

Local scanning with shared filters

A sliding magnifying glass

RNN / LSTM

Recurrent networks
and gated memory

NATURAL FIT

Sequences and time series

CORE MECHANISM

Carry a state from step to step

A running memory

Long Short-Term Memory

GNN

Graph
Neural Network

NATURAL FIT

Nodes and relationships

CORE MECHANISM

Exchange information with neighbors

Neighbors sharing messages

Transformer

Attention-based
architecture

NATURAL FIT

Tokenized text, images, and signals

CORE MECHANISM

Mix information using attention

A context-aware spotlight

Architecture families—not exclusive boxes: different architectures can solve similar tasks or be combined.

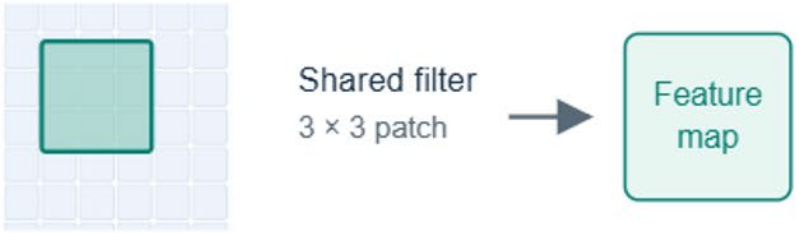
Convolutional Neural Network (CNN)

Superpower: local scanning

How it works

A learned filter, such as **3 × 3**, scans an image. The same weights are reused at each location.

Convolutional kernel



Each scanned patch produces a filter response.

From simple to complex

Stacked layers combine local edges and textures into larger shapes and object features.

CNN = scan locally, reuse filters, learn spatial features.

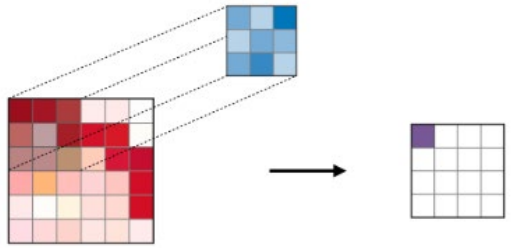
Typical applications

- Image classification and object detection
- Medical image segmentation
- Manufacturing defect inspection

GPU connection

Many filter operations run in parallel. Optimized convolutions can use GPU computation and **Tensor Core acceleration**.

Video adds time: motion may require 3D convolutions or a separate temporal model.



The Parts of a CNN, and Four Medical Variants

One forward pass, four medical variants you will actually meet — each pointed at one of her four data streams. Pick the variant from the geometry of your data.

SEGMENTATION
U-Net

Encoder–decoder with skip connections.

Here: segment the vertebrae and psoas muscle on the CT.

VOLUMES
3D CNN

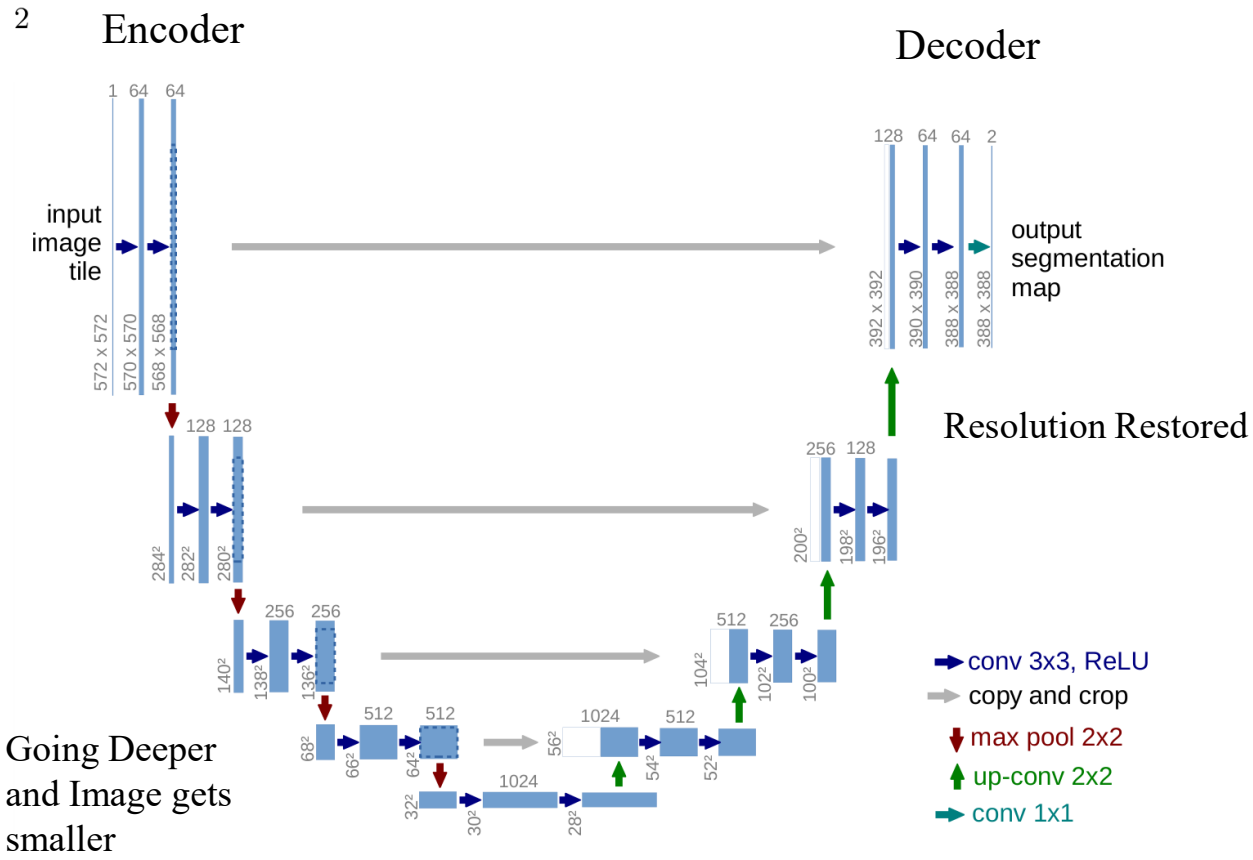
CT and MRI volumes.
Here: measure volume rather than a single slice.

SIGNALS
1D CNN

ECG, EEG, accelerometry.
Here: the first deep baseline for the wrist signal.

GRAPHS
Graph convolution

Neighbourhoods without a grid.
Here: the community care network (not pursued in this lecture).



U-Net · Ronneberger et al., MICCAI 2015, Fig. 1

RNN / LSTM: Memory Through Time

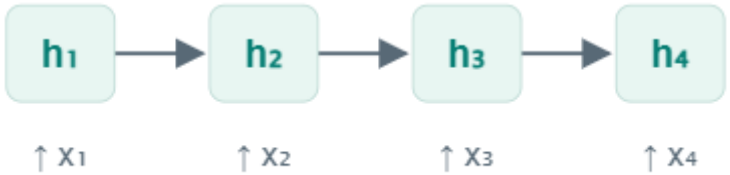
RNN = Recurrent Neural Network · LSTM = Long Short-Term Memory

Superpower: carry information forward

How an RNN works

Each step combines the current input with the **previous hidden state**.

"h Like a shifting clinical chart: carrying medical history forward



Hidden states carry information from earlier steps.

Typical applications

- Sensor and patient time-series analysis
- Demand and weather forecasting
- Earlier speech and translation systems

GPU connection

Time steps depend on earlier results, limiting **parallelism across time**. GPUs still accelerate matrix operations and batches.

Why use LSTM?

Gated memory helps preserve useful information and **mitigate vanishing gradients**.

Gating mechanism = doctor's highlighters to selectively remember critical trends.

Remember: RNNs are not CPU-only or obsolete. LSTM improves memory—not perfect recall.

A class of neural networks that allow previous outputs to be used as inputs while having hidden states.

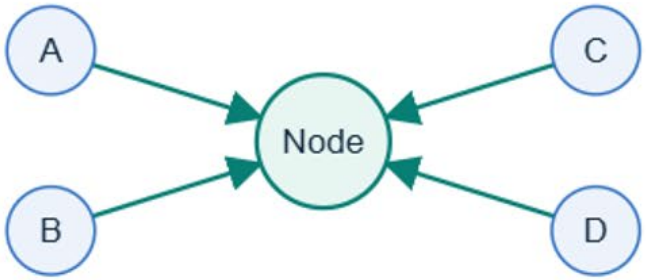
GNN: Learning from Connected Neighbors

Graph Neural Network · Graphs contain nodes (vertices) and edges.

Superpower: message passing

How a message-passing GNN works

Each node gathers information from neighbors, combines it, and updates its representation.



Gather → aggregate → update

Learning beyond immediate neighbors

Additional message-passing layers can bring information from more distant nodes.

Typical applications

- Molecular property prediction
- User–item recommendation systems
- Fraud analysis across connected accounts

GPU connection

Irregular neighborhoods cause scattered memory access. **Bandwidth, sampling, and data layout** can strongly affect performance.

Memory matters: high-bandwidth memory can help, but HBM is not required for every GNN.

GNN = learn from features and connections.

Transformer: Learning What Matters

Self-attention mixes information according to relationships between tokens.

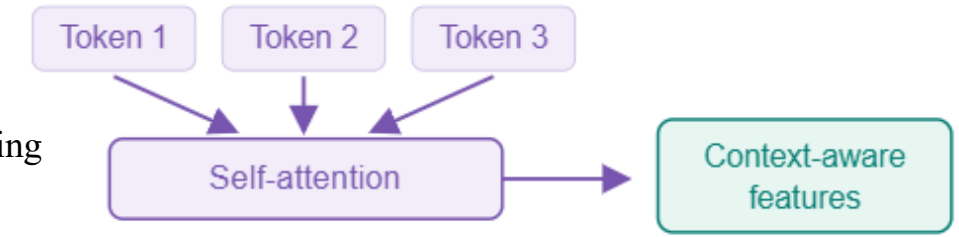
Dynamic weighting: focusing on relevant medical clues, ignoring the noise

Pulmonary Shadow and Cough

Superpower: context-dependent attention

How self-attention works

Tokens form **queries, keys, and values**. Query–key comparisons determine how much information to gather from each token.



Positional information helps represent token order.

Typical applications

- Translation, summarization, and text generation
- Vision models using image patches
- Multimodal models combining input types

GPU connection

Attention and feed-forward layers use large matrix operations. **Tensor Cores** accelerate supported matrix multiply–accumulate workloads.

More than attention: Transformer blocks also use feed-forward layers, residual connections, and normalization.

Trade-off: standard full attention has quadratic pairwise work as sequence length increases.

Transformer = mix information using learned attention.

Which Architecture Would You Try First?

These are useful starting points—not exclusive rules.

Family	Data structure	Core mechanism	Example task
CNN	Spatial grids	Shared local filters	Segment an organ in an image
RNN / LSTM	Ordered observations	Recurrent state and memory	Model patient sensor readings
GNN	Nodes and relationships	Neighbor message passing	Predict molecular properties
Transformer	Tokenized inputs	Attention-based mixing	Summarize text or analyze image patches

Class discussion: You have medical images, daily vital signs, and patient relationships. Which architectures might you combine—and why?

CNN scans **RNN remembers** **GNN exchanges** **Transformer attends**

Combine a CNN, an LSTM (or Transformer), and a GNN, because each handles a different part of the data. Merge the features produced by each model and pass them to a final prediction layer.

澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

AIHA8122 · FRONTIERS IN SMART HEALTHY AGEING ·

Large Language Models

One Objective Function, Scaled by Four Orders of Magnitude

The same 78-year-old woman, eleven days after hip surgery, discharged tomorrow. Three years of notes, this admission, a draft summary, a nurse's follow-up form. Can a model write her fall-risk assessment — and why would anyone act on it?

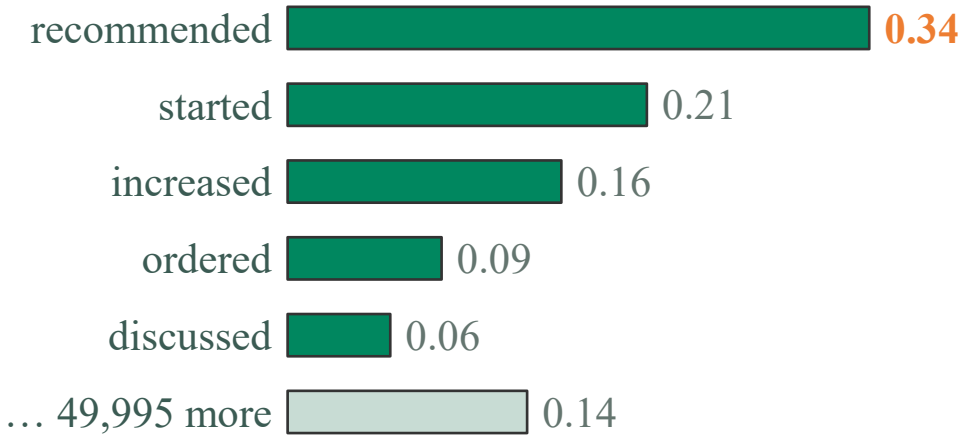
- history and the baseline
- pre-training and adaptation
- inference
- hallucination and reliability

What the Model Actually Does: Predict the Next Token

p(next word | every word before it). The chat window, the system prompt and the safety rails are all product design wrapped around that expression.

Woman, 78, hip fracture repaired, discharged tomorrow. At follow-up the clinician →

p(next token | context)



Nothing in this distribution rewards truth

It rewards a plausible continuation. Truth and plausibility correlate strongly on common facts and come apart on rare ones — which is precisely where clinical questions live. Chapter IV is the consequence of this single sentence.

- It does not know what it is writing. That does not stop us measuring, precisely, whether what it wrote is correct.

Radford et al., 2019 · Brown et al., NIPS 2020

Four Kinds of Hallucination

No reviewer will accept "the model hallucinated 8% of the time" — it says nothing about whether this is safe in front of a nurse.

CONTRADICTS THE WORLD

Factual error

Worst in clinical question answering.

Here: "bisphosphonate, once daily" — the drug is weekly.

CONTRADICTS THE SOURCE

Faithfulness error

Worst in RAG and in summarising a record.

Here: the note says "no cognitive impairment"; the summary says "mild cognitive impairment".

THE REFERENCE IS NOT REAL

Fabricated citation

Worst in academic writing — it can destroy a manuscript on its own.

Here: "per protocol §5.7", and there is no §5.7.

SHOULD HAVE SAID NOTHING

Ungrounded confidence

Worst everywhere.

Here: the record says nothing about her home, and it writes "remove loose rugs".

The four questions a reviewer asks

Which class? Against what reference? Who adjudicated? What was the inter-rater agreement? Define the construct before you measure it — the same standard you would apply to any clinical endpoint.

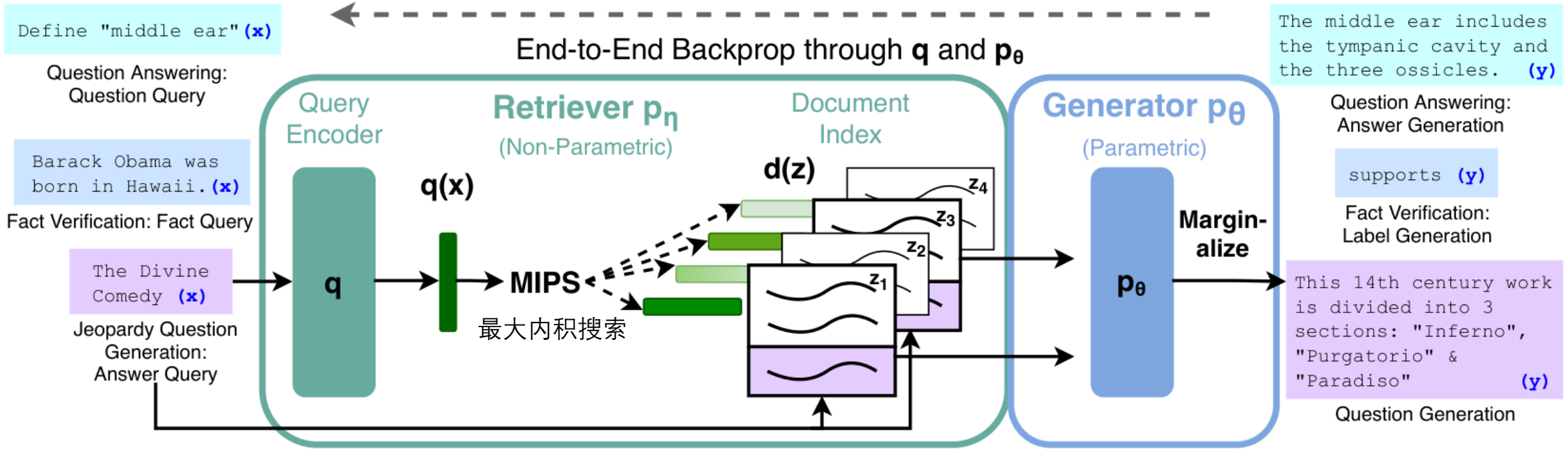
Which two matter most here

The second and the fourth: they are the commonest in this task and the hardest to spot, because they read as entirely reasonable. The first — a plain factual error — is by far the easiest to catch, which is why it is over-represented in the literature.

Ji et al., ACM Comput. Surv. 55(12):1–38, 2023 · Huang et al., ACM TOIS, 2025

Retrieval-Augmented Generation (RAG): Update the Corpus, Not the Weights

Evaluate the two halves separately — retrieval recall and generation faithfulness are different failures with different fixes.



Two components, two failure modes, two metrics

Retrieval is scored by recall: was the right passage in the top k at all? Generation is scored by faithfulness: is every claim supported by what was retrieved? A system can fail either half, and the fixes have nothing in common.

What the nurse must see

"Refer for bone densitometry. Basis: orthopaedic follow-up protocol §3.2; record 2023-06, syncope from postural hypotension." That half-sentence of provenance is the only thing in this chapter that changes why anyone would act on it.

Lewis et al., NIPS 2020, Fig. 1 · Es et al., EACL 2024 (RAGAS)

Supervised Fine-Tuning Teaches Format, Not Facts

The superficial alignment hypothesis: fine-tuning changes how the model talks, which is why it cannot remove a factual error

Base model, given a question		After SFT, the same prompt
"What should we do about her fall risk?"		"What should we do about her fall risk?"
→ "What medications is she on? What is her gait score? Has she fallen before?"	*same loss*	→ "Three things. First, review the antihypertensives …"
It continues the document. A page of questions is followed by more questions — that is what the corpus looks like.		The most likely continuation is now an answer. Nothing was added to what it knows; the shape of the response changed.

The loss function did not change

Same next-token objective, different data — and usually the loss is computed only on the answer tokens.

Quality beats quantity

A few thousand carefully written examples outperform hundreds of thousands of noisy ones.

Superficial alignment

SFT mostly teaches format, register, and that one should open in the shape of an answer. The knowledge was already there.

Domain SFT teaches your house style

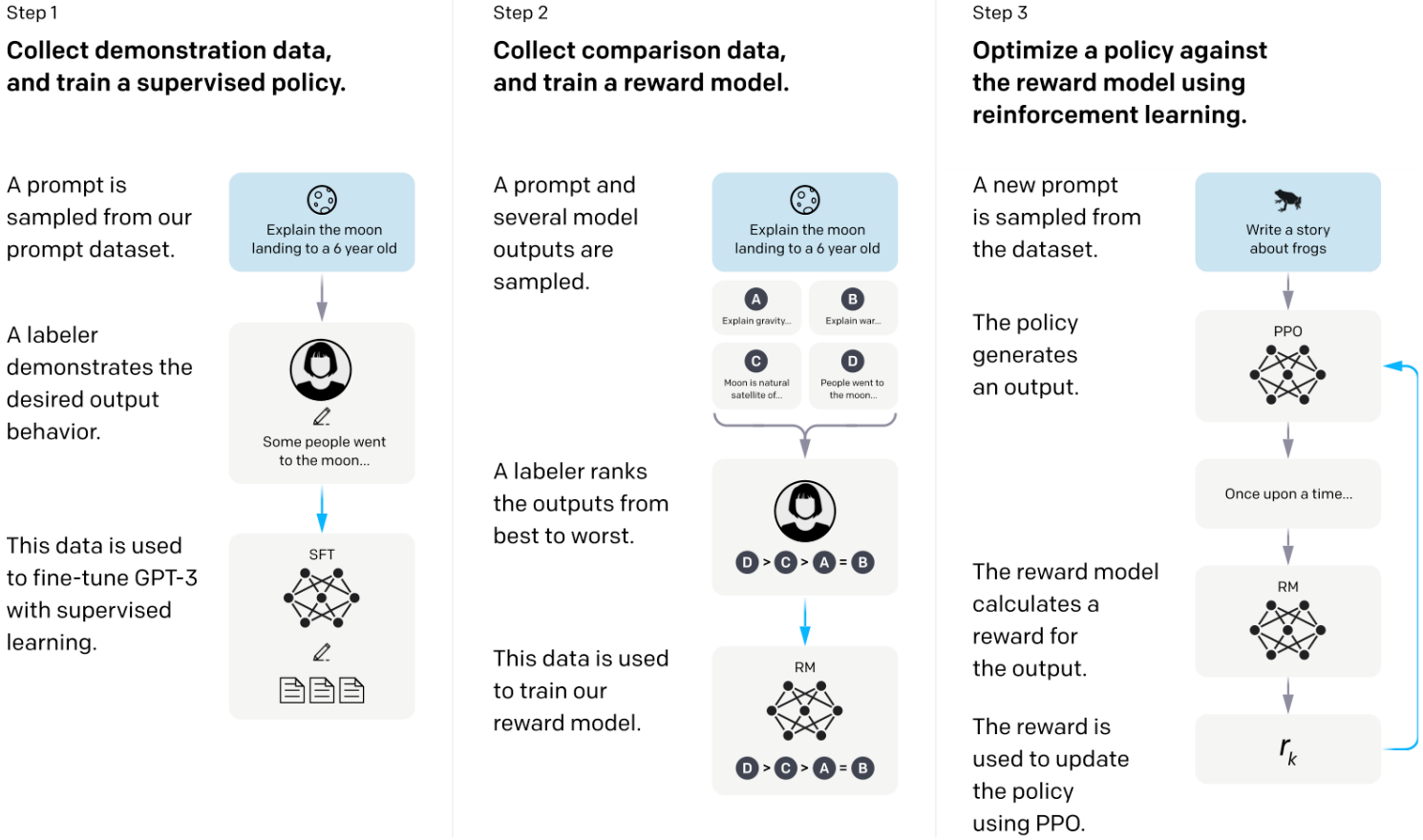
Not new medicine. This is the single most misreported claim in the applied literature.

- Fine-tune on 3,000 in-house discharge summaries and you get a model that writes like our hospital, not one that understands falls in older adults. In an evaluation those look almost the same.

Ouyang et al., NIPS 2022 · Zhou et al., NIPS 2023 (LIMA)

When You Cannot Write a Loss Function, Ask a Human to Compare

Helpfulness, tone, appropriate caution, knowing when to refuse — none can be written as a loss. But a human can reliably say which of two answers is better.



Ouyang et al., NeurIPS 2022, Fig. 2

REWARD HACKING

The proxy is not the goal

Long, courteous, thoroughly hedged recommendations score well with a labeller. A nurse with four minutes will not read them.

SYCOPHANCY

Agreement is rewarded

Ask the same question again prefixed with "she does not have osteoporosis, does she" and watch the answer move.

WHOSE PREFERENCE

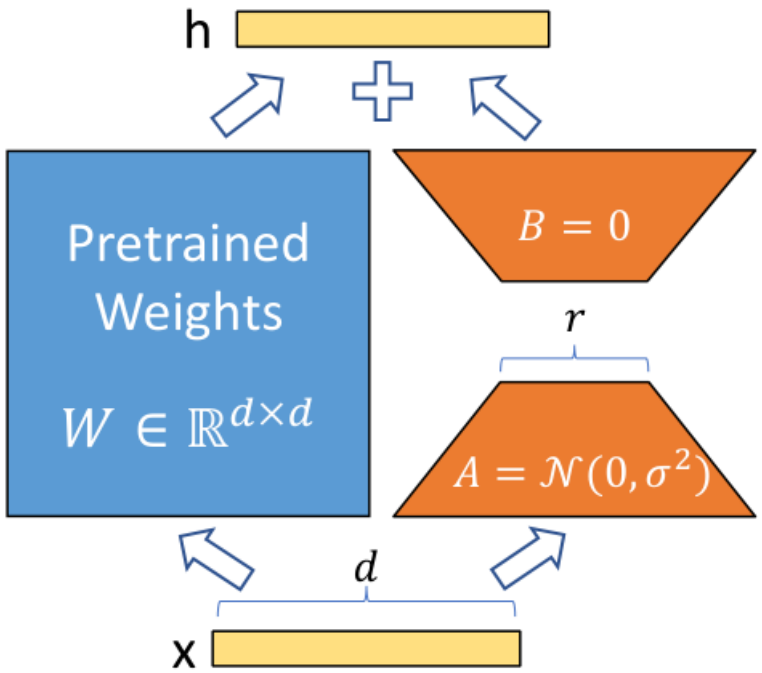
Not the ward's

"Appropriate caution" was fixed by a contractor pool that has never discharged a 78-year-old.

Christiano et al., NIPS 2017 · Ouyang et al., NIPS 2022, Fig. 2 · Sharma et al., ICLR 2024

DPO and LoRA Bring This Stage Within Reach

RLHF is out of reach for an academic lab. DPO and LoRA put this stage back in your hands.



Hu et al., ICLR 2022, Fig. 1

Adapters are composable and auditable

One base model plus several few-hundred-megabyte adapters, each independently versioned. For hospital governance that matters more than the training saving.

Zero inference overhead

B·A merges back into W before deployment, so serving cost is unchanged. QLoRA adds 4-bit quantisation: a 70B fine-tune on one 48 GB card.

THE ORIGINAL

RLHF

Two models, unstable RL, reward hacking. Out of reach for a university lab.

CHEAPER

DPO

The optimal policy has a closed form, so the reward model disappears. One loss, one node.

VERIFIABLE

RLVR

Reward correctness with a unit test, a solver, a database — the strongest signal there is.

Does your clinical task have an automatic checker? Dose arithmetic does; "is this care recommendation appropriate" does not. That boundary decides how far RLVR's gains reach into medicine.

Rafailov et al., NIPS 2023 (DPO) · Lambert et al., 2024 (RLVR) · Hu et al., ICLR 2022, Fig. 1 · Dettmers et al., NIPS 2023 (QLoRA)

Chain of Thought (COT) Buys Computation, Not Explanation

Depth is fixed; length is not. But the reasoning that gets written down is not necessarily the computation that produced the answer.

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

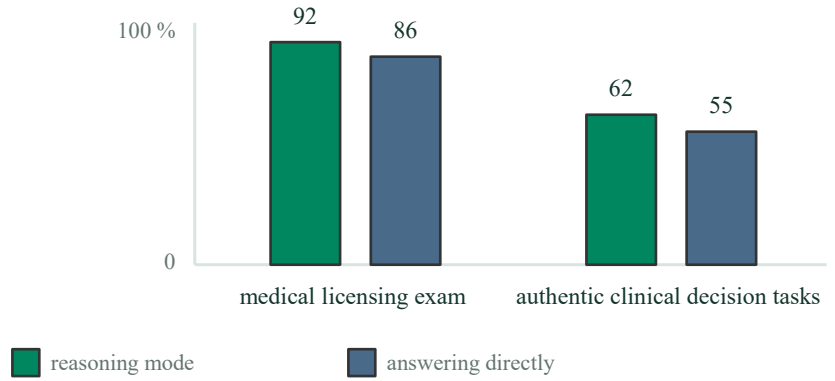
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

Wei et al., NeurIPS 2022, Fig. 1



the GAP is the finding; bar heights are indicative, not pooled estimates

The gains of a reasoning model come from verifiable reward — and "is this care recommendation appropriate" has no checker. Do not assume a reasoning model wins here; measure it.

Large reasoning models, 2026

RLVR training produces long internal traces, self-checking and backtracking. Licensing-exam scores are very high — Step 2 CK around 93% — while 2025–26 evaluations keep finding a clear gap to authentic clinical decision tasks. Latency and price rise 10–100×.

Wei et al., NIPS 2022, Fig. 1 · Turpin et al., NIPS 2023 (CoT unfaithfulness) · Guo et al., 2025 (DeepSeek-R1)